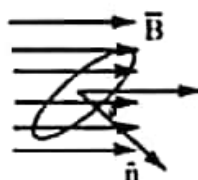


3. ELECTROMAGNETIC INDUCTION

SYNOPSIS

1. Magnetic flux :

- i) The magnetic flux through a surface (of area $\vec{A}$) placed in a uniform magnetic field $\vec{B}$ is defined as $\phi = \vec{B} \cdot \vec{A} = BA \cos \theta$
Where θ is the angle between $\vec{B}$ and $\vec{A}$



- ii) Direction of $\vec{A}$ is in the direction of unit vector ($\hat{n}$) drawn perpendicular to the plane of the surface

2. Faraday's Laws :

- i) Whenever there is a change of magnetic flux through a circuit, there will be an induced emf and this will last as long as the change persists
ii) The magnitude of induced emf is equal to the time rate of change of magnetic flux

$$\text{Mathematically, } e = \frac{d\phi}{dt}$$

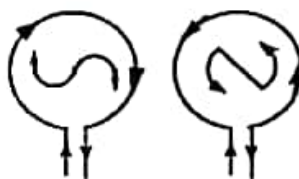
The negative sign indicates that the emf is induced so as to oppose the change of flux .

3. Lenz's Law :

- i) "The polarity of the induced emf is such that it tends to produce a current which opposes the change that produces it"

Lenz's law is usually understood in terms of closed circuits and induced currents. If the circuit is open, we should think in terms of what would happen if it were closed and thus determine the polarity of induced emf.

- ii) **Poles of a coil can be found.** If the current is clockwise the pole will be south, if the current is anti-clockwise, its pole will be north as in figure.



- iii) Lenz's law is a consequence of law of conservation of energy.
iv) As $\phi = BAN \cos \theta$,
The change in flux can be caused by changing B (or) A (or) N (or) θ
v) Average induced emf when only 'B' is changed is given by

$$e = -AN \cos \theta \frac{(B_2 - B_1)}{(t_2 - t_1)}$$

Where B_1 & B_2 are the magnetic fields at instants t_1 & t_2 respectively .

If the plane of the coil perpendicular to magnetic field, $\theta = 0^\circ$

$$\text{then } e = -AN \frac{(B_2 - B_1)}{(t_2 - t_1)}$$

Instantaneous emf is given by

$$e = -AN \cos \theta \frac{dB}{dt}$$

vi) Average induced emf when only 'A' is changed is given by

$$e = -BN \cos \theta \frac{(A_2 - A_1)}{(t_2 - t_1)}$$

Where A_1 & A_2 are the areas of the coils at the instants t_1 & t_2 respectively. Instantaneous emf is given by

$$e = -BN \cos \theta \frac{dA}{dt}$$

vii) Average induced emf when only ' θ ' is changed is given by

$$e = -BAN \frac{(\cos \theta_2 - \cos \theta_1)}{(t_2 - t_1)}$$

Where θ_1 & θ_2 are the angles made by $\vec{A}$ with $\vec{B}$. Instantaneous emf is given by

$$e = -BAN \frac{d(\cos \theta)}{dt}$$

If the coil is rotated with constant angular velocity ' ω ' then

$$e = -BAN \frac{d(\cos \omega t)}{dt} = BAN \omega \sin \omega t$$

$$\therefore e = BAN \omega \sin \omega t$$

If $\omega t = 90^\circ$, the plane is parallel to magnetic field and then induced emf is maximum.

$$\text{Then } e_0 = BAN \omega$$

$$\therefore e = e_0 \sin \omega t$$

This is the principle in AC generator.

viii) An ac generator converts mechanical energy into electrical energy. The device used for the purpose is called ac generator.

ix) The induced emf depends upon (i) strength of the magnetic field, (ii) area of the coil, (iii) speed of rotation, and (iv) the number of turns of the coil.

x) If the generator connected to a circuit of resistance ' R ', current through the circuit

$$\text{is } i = \frac{e}{R} = \frac{BAN \omega}{R} \sin \omega t$$

xi) Power delivered to the circuit is $P = ei$

xii) The flux change by changing the number of turns is practically difficult.

xiii) Current induced in a coil is given by $i = \frac{e}{R} = -\frac{1}{R} \frac{\Delta \Phi}{\Delta t}$ But $i = \frac{q}{\Delta t} \therefore q = -\frac{\Delta \Phi}{R}$

i.e. Induced charge is independent of the time interval in which the change of flux is caused.

4. Motional EMF

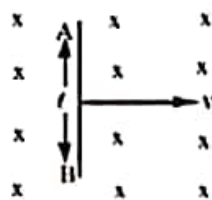
- i) When a rod of length ' l ' is moved with a velocity ' v ' perpendicular to its length in a uniform magnetic field B which is perpendicular to both its length as well as its velocity, then emf induced across its ends is given by $\epsilon = Blv$

- ii) If the rod is moved making angle ' θ ' with its length, then $\epsilon = Blv \sin \theta$

In vector form $\epsilon = \vec{B} \cdot (\vec{l} \times \vec{v})$

- iii) Fleming's right hand rule is used to find the polarity of emf (or) current in the rod if the rod is connected to an external resistance.

Stretch the thumb, fore finger and middle finger such that they are mutually perpendicular and if thumb is in the direction of velocity, forefinger is in the direction of field, then middle finger gives the direction of induced current.



- iv) In the above shown diagram, if the ends A and B of a rod are connected by a conducting wire, then current flows from B to A inside the rod. As the rod acts as a cell, and inside the cell the current flows from negative terminal to positive terminal, here B becomes -ve and A becomes +ve end.

- v) If ends A and B are connected by an external resistance R , then current in the rod is given by

$$i = \frac{Blv}{R}$$

- vi) Now, the force acting on the rod is given by $F = Bil$

$$B \left(\frac{Blv}{R} \right) l = \frac{B^2 l^2 v}{R}$$

- vii) That is, we should apply a force of $\frac{B^2 l^2 v}{R}$ on the rod, to move it with constant velocity.

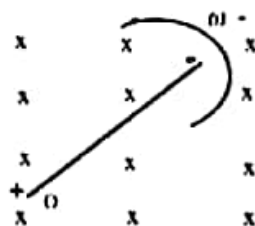
- viii) The power applied by external agent in moving the rod with constant velocity is given by

$$P = Fv = \frac{B^2 l^2 v^2}{R}$$

- ix) If an irregular shaped conductor was moved in a uniform magnetic field, take effective length (vectorial sum of small current elements) of conductor to find emf. $\epsilon = B(l_{\text{eff}})v$

- x) If a conducting disc of radius ' R ' rotates with constant angular velocity ' ω ' in a uniform magnetic field, which is perpendicular to the plane of disc, the emf between centre & rim is $\epsilon = \frac{1}{2} BR^2 \omega$

- xi) If a rod of length ' l ' is rotated with a constant angular velocity ' ω ' about an axis passing through its end (O) and perpendicular to its length and if a uniform magnetic field $\vec{B}$ is present perpendicular to it, then emf across its ends is given by $\epsilon = \frac{1}{2} B l^2 \omega$



5. Induced electric field :

- A time varying magnetic flux induces an emf and a current in conducting loop.
- The flow of charges in the loop is due to an electric field in it, called induced electric field, due to varying magnetic field.

$$\text{iii) } \int \vec{E} \cdot d\vec{r} = \frac{d\phi}{dt} \rightarrow \int \vec{E} \cdot d\vec{r} = \Lambda \left(\frac{d\phi}{dt} \right)$$

- Induced electric field are non-conservative fields and these lines of forces form closed loops.
- Induced electric field due to varying magnetic fields established even in the absence of any material objects, called vortex field (or) space electric field.

Eddy Currents

- When bulk pieces of conductors are subjected to changing magnetic flux, induced currents are produced in them.
- The flow patterns of induced currents resemble the swirling eddies in water. This effect was discovered by Foucault and these currents are called *eddy currents (or) Foucault currents*.
- A copper plate is allowed to swing like a simple pendulum between the pole pieces of a strong magnet, its motion is damped and the plate comes to rest in the magnetic field due to eddy currents in the plate.

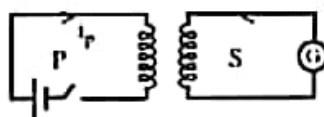
Advantages :

Eddy currents are used in

- Magnetic braking in trains;
- Electromagnetic damping;
- Induction furnace;
- Electric power meters;

6. Mutual Induction :

- When current in one coil changes, magnetic flux linked with the second coil placed near by it also changes. Then emf induced in secondary is called mutually induced emf and the phenomenon is called mutual induction.



- If ' i_p ' is current flowing in the primary coil, ' ϕ_s ' is magnetic flux linked with secondary coil, then

$$\phi_s \propto i_p$$

LECTURE SHEET

EXERCISE-I

(Magnetic flux, Faraday's laws)

LEVEL-I (MAIN)

Straight Objective Type Questions

Magnetic flux, Faraday's law, Lenz's law :

- A coil of 100 turns and area 5 square centimetre is placed in a magnetic field $B = 0.2T$. The normal to the plane of the coil makes an angle of 60° with the direction of the magnetic field. The magnetic flux linked with the coil is
 1) $5 \times 10^{-3} \text{ Wb}$ 2) $5 \times 10^{-5} \text{ Wb}$ 3) 10^{-2} Wb 4) 10^{-4} Wb
- A magnetic field of $2 \times 10^{-2} T$ acts at right angles to a coil of area 100 cm^2 with 50 turns. If the average emf induced in the coil is $0.1V$ when it is removed from the field in time t . The value of t is
 1) $0.1s$ 2) $0.01s$ 3) $1s$ 4) $20s$
- The magnetic flux of $500 \mu\text{Wb}$ passing through a 200 turn coil is reversed in 20×10^{-3} seconds. The average e.m.f. induced in the coil in volt is
 1) 2.5 2) 5.0 3) 7.5 4) 10.0
- A rectangular coil of 20 turns and area of cross section 25 cm^2 has a resistance of 100 ohm. If the magnetic field which is perpendicular to the plane of coil changes at the rate of 1000 tesla/sec, the current in the coil is
 1) 1 amp 2) 0.5 amp 3) 5 amp 4) 50 amp
- In a magnetic field of $0.05T$, area of a coil changes from 101 cm^2 to 100 cm^2 without changing the resistance which is 2Ω . The amount of charge that flow during this period is
 1) $2.5 \times 10^{-6} C$ 2) $2 \times 10^{-6} C$ 3) $10^{-6} C$ 4) $8 \times 10^{-6} C$

Numerical Type Questions

- The number of turns in the coil of an ac generator is 5000 and the area of the coil is 0.25 m^2 . The coil is rotated at the rate of 100 cycles/sec in a magnetic field of $0.2 T$. The peak value of the emf (in kV) generated is
- A circular coil of area 8 m^2 and number of turns 20 is placed in a magnetic field of $2T$ with its plane perpendicular to it. It is rotated with an angular velocity of 20 rev/s about its natural axis. The emf induced is
- A coil having 500 square loops each of side 10 cm is placed normal to a magnetic flux which increases at the rate of 1.0 tesla/second . The induced e.m.f. in volts is
- A coil of 40Ω resistance has 100 turns and radius 6 mm is connected to ammeter of resistance of 160 ohms . Coil is placed perpendicular to the magnetic field. When coil is taken out of the field, $32 \mu C$ charge flows through it. The magnetic induction field strength (in Tesla) is
- Three identical square loops are made of a single wire as shown.

The magnetic field depending on time as $B = B_0 t$ is present in the region as shown. The side of the square is a and resistance per unit length of the wire is λ . The induced current in the loop is $\frac{B_0 a}{4k\lambda}$ where $k = \underline{\hspace{1cm}}$



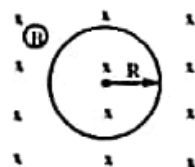
LEVEL-II (ADVANCED)

Straight Objective Type Questions

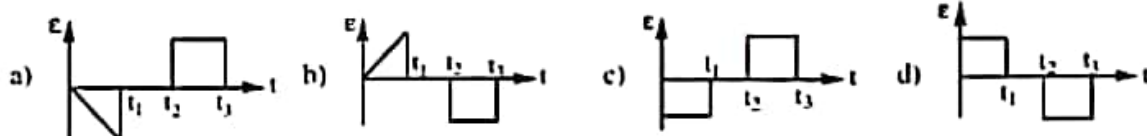
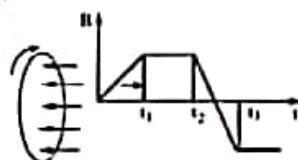
Magnetic flux, Faraday's laws of EMI

1. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of ring. If radius R varies as a function of time t as $R = R_0 + t^2$. The emf induced in the loop is

- a) $2\pi Bt(R_0 + t^2)$ clockwise
 b) $4\pi Bt^2(t^2 + R_0)$ anticlockwise
 c) $4\pi Bt(t^2 + R_0)$ anticlockwise
 d) $4\pi Bt(t^2 + R_0)$ clockwise

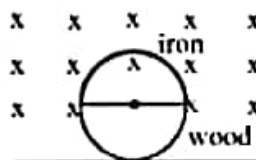


2. A wire loop is placed in a region of time varying magnetic field which is oriented orthogonally to the plane of the loop as shown in the figure. The graph shows the magnetic field variation as the function of time. Assume the positive emf is the one which drives a current in the clockwise direction and seen by the observer in the direction of B . Which of the following graphs best represents the induced emf as a function of time.



3. Figure shows a circular wheel of radius 10.0 cm whose upper half, is made of iron and the lower half of wood. The two junctions are joined by an iron rod. A uniform magnetic field B of magnitude 2.00×10^{-4} T exists in the space above the central line as suggested by the figure. The wheel is set into pure rolling on the horizontal surface. If it takes 2.00 seconds for the iron part to come down and the wooden part to go up, find the average emf induced during this period.

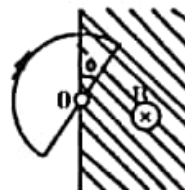
- a) 1.25×10^{-7} V
 b) 1.57×10^{-6} V
 c) 5.57×10^{-6} V
 d) 3.57×10^{-7} V



4. A square loop of side l_0 is placed on a smooth horizontal surface, where a uniform, vertical magnetic field of constant magnitude B_0 exists. If the temperature of the surroundings changes as $\theta = a + bt$, where a and b are positive constants of appropriate dimensions, t is instant of time. The magnitude of emf linked with the loop at any instant t is (α is the coefficient of linear expansion of the material of the wire)

- a) $B_0 l_0 b \alpha [a + b \alpha t]$
 b) $2 B_0 l_0^2 b \alpha [1 + b \alpha t]$
 c) $B_0 l_0^2 \alpha [a + b \alpha t]$
 d) $B_0 l_0^2 a \alpha [1 + b \alpha t]$

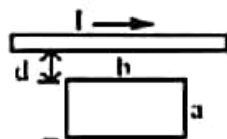
5. A wire loop enclosing a semi-circle of radius a is located on the boundary of a uniform magnetic field of induction B . At the moment $t = 0$, the loop is set into rotation with a constant angular acceleration α about an axis O coinciding with a line of vector B on the boundary. Find the emf induced in the loop as a function



of time t . The arrow in the figure shows the emf direction taken to be positive.

- a) $\alpha Ba^2 t$ b) $\frac{\alpha Ba^2 t}{4}$ c) $\frac{1}{2} \alpha Ba^2 t$ d) $2 \alpha Ba^2 t$

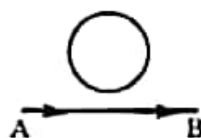
6. A long straight wire is parallel to one edge as shown in fig. If the current in the long wire varies with time as $I = I_0 e^{-t/\tau}$, what will be the induced emf in the loop?



- a) $\frac{\mu_0 b I}{\pi \tau} \ln\left(\frac{d+a}{d}\right)$ b) $\frac{\mu_0 b I}{2\pi \tau} \ln\left(\frac{d+a}{d}\right)$ c) $\frac{2\mu_0 b I}{\pi \tau} \ln\left(\frac{d+a}{d}\right)$ d) $\frac{\mu_0 b I}{\pi \tau} \ln\left(\frac{d}{d+a}\right)$

7. In the arrangement shown in given figure current from A to B is increasing in magnitude. Induced current in the loop will

- a) have clockwise direction
b) have anticlockwise direction
c) be zero
d) oscillate between clockwise and anticlockwise



8. Magnetic flux linked with a stationary loop of resistance 4Ω varies with respect to time during the time period 2 second, as follows: $\Phi = 3t(2-t)$. Find the amount of heat generated (in joules) during that time.

- a) 6 b) 8 c) 2 d) 1

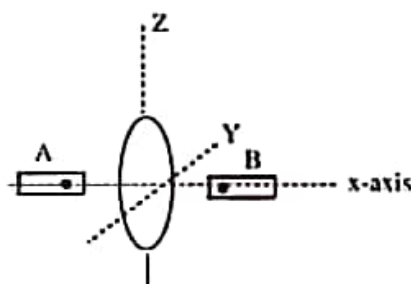
More than One correct answer Type Questions

9. A loop is fixed with its centre at origin, and lies in the yz plane. Two identical thin bar magnets are placed on x -axis at symmetric distances from the fixed loop as shown. Dot indicates the north-pole of the magnet. Imagine the following situations.

Situation-1: The two magnets move towards the origin with equal uniform speeds

Situation-2 : The two magnets move towards the origin with unequal uniform speeds

Select the correct option(s) before any of the magnet crosses the origin.



- a) The net induced current in the fixed loop is zero due to situation - 1
b) The net induced current in the loop is non-zero due to situation - 1
c) The net induced current in the fixed loop is zero due to situation - 2
d) The net induced current in the fixed loop is non-zero due to situation - 2

Linked Comprehension Type Questions

Passage - I :

Two circular concentric conducting coils A and B have radii r, R (r being very small and $r \ll R$) and number of turns $n, 2n$ respectively. Their planes are inclined at 60° to each other. Now answer following parts

10. If coil A carries current i then Magnetic flux through coil B is

- a) $\frac{\mu_0 \pi n^2 R^2 i}{2r}$ b) $\frac{\mu_0 \pi n^2 r^2 i}{2R}$ c) not possible to determine d) none of these

11. Amount of charge that will flow through any cross section of coil B (of resistance K) when current i in coil A is reversed in direction only will be

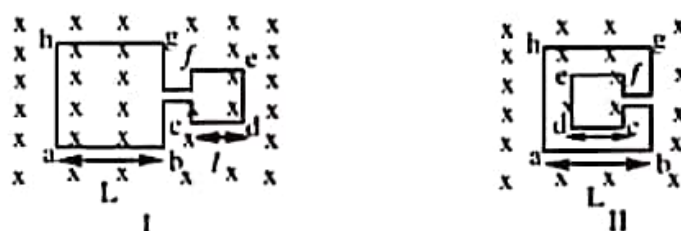
- a) $\frac{\mu_0 \pi n^2 R^2 i}{K r}$ b) $\frac{\mu_0 \pi n^2 R^2}{4 K r}$ c) $\frac{\mu_0 n r^2}{2 K R}$ d) $\frac{\mu_0 n^2 r}{4 K}$

12. If both the coils carry same current i then modulus of work done by an external agent in slowly rotating coil A by 180° about coinciding diameter of the coils is

- a) $\frac{\mu_0^2 \pi n^4 r^2 i}{2 R^2}$ b) $\frac{\mu_0^2 \pi n^4 i r^3}{3 R^2}$ c) $\frac{\mu_0 \pi n^2 i^2 r^2}{R}$ d) none of these.

Passage - II :

The adjoining figure shows two different arrangements in which two square wire frames are placed in a uniform constantly decreasing magnetic field B .



13. The value of magnetic flux in each case is given by

- a) Case I : $\phi = \pi (L^2 + l^2)B$; Case II: $\phi = \pi (L^2 - l^2)B$
 b) Case I: $\phi = \pi (L^2 + l^2)B$; Case II: $\phi = \pi (L^2 + l^2)B$
 c) Case I: $\phi = (L^2 + l^2)B$; Case II: $\phi = (L^2 - l^2)B$
 d) Case I: $\phi = (L + l)^2 B$; Case II: $\phi = \pi (L - l)^2 B$

14. The direction of induced current in the case I is

- a) from a to b and from e to d b) from a to b and from f to e
 c) from b to a and from d to c d) from b to a and from e to f

15. The direction of induced current in the case II is

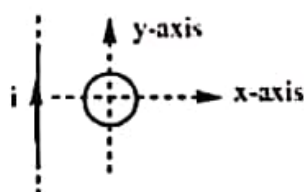
- a) from a to b and from e to d b) from b to a and from f to e
 c) from b to a and from e to d d) from a to b and from d to c

16. If I_1 and I_2 are the magnitudes of induced current in the cases I and II, respectively, then

- a) $I_1 = I_2$ b) $I_1 > I_2$ c) $I_1 < I_2$ d) nothing can be said

Matrix Matching Type Questions

17. A conducting circular rigid loop near a long straight current carrying wire as shown. Match the following table :

**COLUMN - I**

- A) If current is increased
 B) If current is decreased
 C) If wire is moved away from the loop maintaining constant current
 D) If wire is moved towards the wire maintaining constant current

COLUMN - II

- p) Induced current in the loop is clockwise
 q) Induced current in the loop is anticlockwise
 r) Wire will attract the loop and there will be a torque about y-axis
 s) Wire will repel the loop and there will be no torque about y-axis

18. Match the following

COLUMN - I

- A) A metallic loop rotating about an axis parallel to the magnetic field
 B) Lenz's law explains
 C) If a current in the loop decreases
 D) Fleming's rule explains

COLUMN - II

- p) Conservation of energy
 q) Direction of magnetic field
 r) No emf is induced in the loop
 s) emf is induced in the loop

EXERCISE-II*(Motional emf)***LEVEL-I (MAIN)**Straight Objective Type Questions

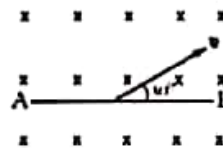
- The wing span of an aeroplane is 20 metre. It is flying in a field, where the vertical component of magnetic field of earth is 5×10^{-5} tesla, with velocity 360 km/h. The potential difference produced between the blades will be
 1) 0.10 V 2) 0.15 V 3) 0.20 V 4) 0.30 V
- A copper disc of radius 0.1 m is rotated about its natural axis with 10 rps in a uniform magnetic field of 0.1 T with its plane perpendicular to the field. The emf induced across the radius of the disc is
 1) $\pi \times 10$ V 2) $2\pi \times 10$ V 3) $\pi \times 10^{-2}$ V 4) $2\pi \times 10^{-2}$ V
- A wheel with 10 metallic spokes each 0.5m long is rotated with a speed of 120 rev/min in a plane normal to the earth's magnetic field at the place. If the magnitude of the field is 0.4 G, the induced e.m.f. between the axle and the rim of the wheel equal to
 1) 1.256×10^{-3} V 2) 6.28×10^{-4} V 3) 1.256×10^{-4} V 4) 6.28×10^{-5} V
- A vertical ring of radius r and resistance R falls vertically. It is in contact with two vertical rails which are joined at the top. The rails are without friction and resistance. There is a horizontal uniform magnetic field of magnitude B perpendicular to the plane of the ring and the rails. When the speed of the ring is v , the current in the section PQ is

- 1) zero 2) $\frac{2Brv}{R}$ 3) $\frac{4Brv}{R}$ 4) $\frac{8Brv}{R}$



5. A conducting rod AB of length $l = 1$ m is moving at a velocity $v = 4$ m/s making an angle 30° with its length. A uniform magnetic field $B = 2$ T exists in a direction perpendicular to the plane of motion. Then

- 1) $V_A - V_B = 8$ V 2) $V_A - V_B = 4$ V
3) $V_B - V_A = 8$ V 4) $V_B - V_A = 4$ V

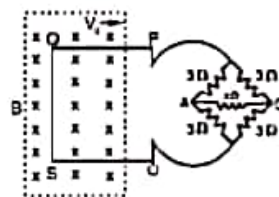


6. A copper rod of mass m slides under gravity on two smooth parallel rails l distance apart and set at an angle θ to the horizontal. At the bottom, the rails are joined by a resistance R . There is a uniform magnetic field perpendicular to the plane of the rails. The terminal velocity of the rod is

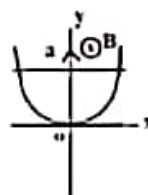
- 1) $\frac{mgR \cos \theta}{B^2 l^2}$ 2) $\frac{mgR \sin \theta}{B^2 l^2}$ 3) $\frac{mgR \tan \theta}{B^2 l^2}$ 4) $\frac{mgR \cot \theta}{B^2 l^2}$

Numerical Type Questions

7. A square metal wire loop of side 10 cm and resistance 1Ω is moved with a constant velocity v_0 in a uniform magnetic field of induction $B = 2$ weber/m² as shown in the figure. The magnetic field lines are perpendicular to the plane of the loop (directed into the paper). The loop is connected to a network of resistors each of value 3Ω . The resistances of the lead wires OS and PQ are negligible. What should be the speed in (m/s) of the loop so as to have a steady current of 1 mA in the loop?



8. A uniform magnetic field exists in region given by $\vec{B} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. A rod of length 5 m is placed along y -axis is moved along x -axis with constant speed 1 m/sec. Then induced e.m.f. in the rod in volts is:
9. A wire is bent as a parabola $y = cx^2$ is located in a uniform magnetic field of magnitude B perpendicular to the xy plane as shown. At the instant $t = 0$ a long metal rod starts translating from the vertex of the parabola with a constant acceleration a along positive y -axis. The induced emf across the points of contact of the rod with the wire is given by $\mathcal{E} = 2By\sqrt{\frac{ka}{c}}$; where $k = \underline{\hspace{1cm}}$.



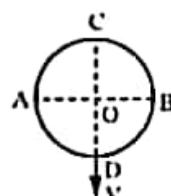
10. A 0.1 m long conductor carrying a current of 50 A is perpendicular to a magnetic field of 1.25 mT. The mechanical power to move the conductor with a speed of 1 m/s⁻¹ is mW.
11. A conducting bar of mass m and length l moves on two frictionless rails in the presence of a uniform magnetic field B_0 directed perpendicular to the plane of parallel rails. The rails are connected to a resistance R . The bar is given velocity V_0 to the right at $t=0$, the velocity of bar after ' t ' sec is given by $V = V_0 e^{-(B^2 l^2 t / kmR)}$. The value of ' k ' is

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. A vertical rod of length l is moved with constant velocity v towards East. The vertical component of the earth's magnetic field is B and the angle of dip is θ . The induced e.m.f. in the rod is:
 a) $B / v \cot \theta$ b) $B / v \sin \theta$ c) $B / v \tan \theta$ d) $B / v \cos \theta$
2. A vertical conducting ring of radius R falls vertically with a speed V in a horizontal uniform magnetic field B which is perpendicular to the plane of the ring :

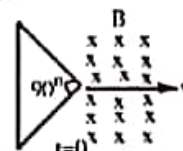
- a) A and B are at same potential
 b) C and D are at same potential
 c) current flows in clockwise direction
 d) current flows in anticlockwise direction



3. A square loop of area $2.5 \times 10^{-1} \text{ m}^2$ and having 100 turns with a total resistance of 100Ω is moved out of a uniform magnetic field of 0.40 T in 1 sec with a constant speed. Then work done, in pulling the loop is

- a) zero b) 1 mJ c) 1 μJ d) 0.1 mJ

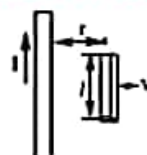
4. The figure shows an isosceles triangle wire frame with apex angle equal to $\frac{\pi}{3}$. The frame starts entering into the region of uniform magnetic field B with constant velocity v at $t = 0$. The longest side of the frame is perpendicular to the direction of velocity. If i is the instantaneous current through the frame then choose the alternative showing the correct variation of i with time.



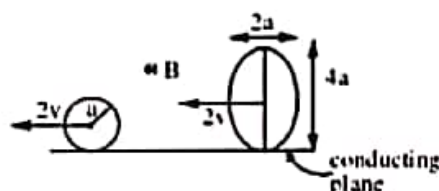
- a) b) c) d)

5. A conducting rod moves with constant velocity v perpendicular to the long, straight wire carrying a current I as shown compute that the emf generated between the ends of the rod.

- a) $\frac{\mu_0 v I l}{\pi r}$ b) $\frac{\mu_0 v I l}{2\pi r}$
 c) $\frac{2\mu_0 v I l}{\pi r}$ d) $\frac{\mu_0 v I l}{4\pi r}$



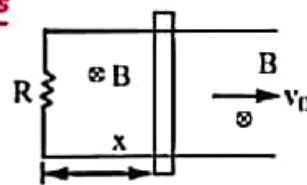
6. A conducting circular ring and a conducting elliptical ring both are moving pure translationally in a uniform magnetic field of strength B as shown in figure on a horizontal conducting plane then potential difference between top most points of circle and ellipse is :



- a) $12 va$ b) $4 vBa$ c) $8 vBa$ d) $2 vBa$

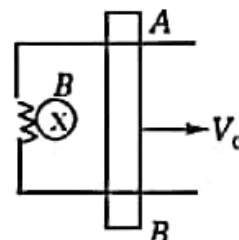
More than One correct answer Type Questions

13. A conducting rod of length l is moved at constant velocity v_0 on two parallel conducting, smooth, fixed rails, that are placed in a uniform constant magnetic field B perpendicular to the plane of the rails as shown in figure. A resistance R is connected between the two ends of the rail. Then which of the following is/are correct?
- a) The thermal power dissipated in the resistor is equal to rate of work done by external person pulling the rod.
- b) If applied external force is doubled then a part of external power increases the velocity of rod.
- c) Lenz's law is not satisfied for direction of current in Loop.
- d) If resistance R is doubled then power required to maintain the constant velocity v_0 becomes half.



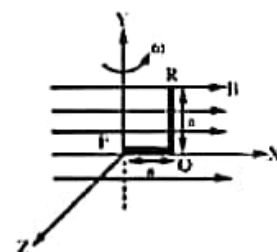
14. A conducting rod AB of mass ' m ' slides without friction over two long conducting horizontal rails separated by a distance ' l '. At the left rails are inter connected by a resistor R . The system is located in uniform magnetic field acting perpendicular to the plane of the loop. At the moment $t = 0$, the rod AB starts moving to the right with an initial velocity V_0 . Neglect resistance of rails and self induction of the circuit

- a) The rod covers a distance $\frac{mRV_0}{2B^2l^2}$ before stopping
- b) The rod covers a distance $\frac{mRV_0}{B^2l^2}$ before stopping
- c) The heat liberated in the resistor during motion of the rod is $\frac{1}{2}mV_0^2$
- d) The heat liberated in the resistor during motion of the rod is $\frac{1}{4}mV_0^2$



15. In a region there exist a magnetic field B_0 along positive x -axis. A metallic wire of length $2a$ and one side along x -axis and one side parallel of y -axis is rotating about y -axis with a angular velocity ω . Then at the instant shown.

- a) Potential difference across PQ is zero
- b) Potential difference across PQ is $\frac{1}{2}B_0\omega a^2$
- c) Potential difference across QR is $\frac{1}{2}B_0\omega a^2$
- d) Potential difference across QR is $B_0\omega a^2$



16. A flexible wire loop in the shape of a circle has a radius that grows linearly with time, as $r = r_0 t$. There exists a time varying magnetic field in the region as $B = B_0 t$, where B_0 and r_0 are positive constants with appropriate dimensions

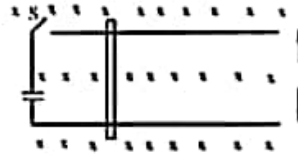
- a) The rate of change of induced emf varies linearly with time
- b) The induced emf varies parabolically with time
- c) The dimensional formula of $\frac{B_0}{r_0}$ is $M^1L^{-1}T^{-2}A^{-1}$
- d) The dimensions of $\frac{B_0}{r_0}$ is independent of the physical quantity length

Linked Comprehension Type Questions

Passage :

One end of a horizontal track of separation l and negligible resistance is connected to a capacitor of capacitance C charged to voltage V_0 . The inductance of the assembly is negligible. The system

is placed in a homogeneous, vertical magnetic field of induction B , as shown in the figure. A frictionless conducting rod of mass m and resistance R is placed perpendicularly onto the track. The polarity of the capacitor is such that the rod is repelled from the capacitor when the switch S is closed.



17. The maximum velocity of the rod $V_{\max}$ is

a) $\frac{B l C V_0}{m + B^2 l^2 C}$

b) $\frac{V_0}{B l}$

c) $\frac{V_0}{m}$

d) $\frac{2 B l C V_0}{2 m + B^2 l^2 C}$

18. The minimum charge on the capacitor $Q_{\min}$ is

a) zero

b) $\frac{C V_0}{2}$

c) $\frac{B^2 l^2 C^2 V_0}{m + B^2 l^2 C}$

d) $\frac{2 B^2 l^2 C^2 V_0}{m + 2 B^2 l^2 C}$

19. The heat generated in the system is

a) zero

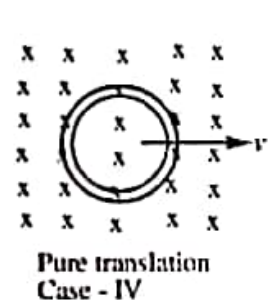
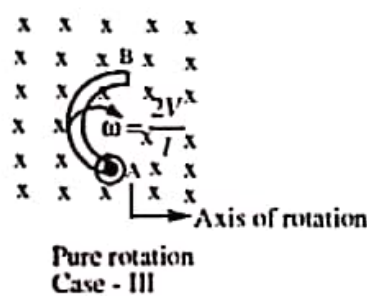
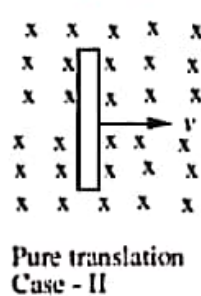
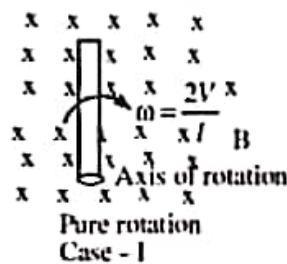
b) $\frac{1}{2} C V_0^2 - \frac{Q_{\min}^2}{2 C}$

c) $\frac{1}{2} C V_0^2 - \frac{1}{2} m V_{\max}^2 - \frac{Q_{\min}^2}{2 C}$

d) none of these

Matrix Matching Type Questions

20. Consider four cases in which conducting wire element is moved with constant velocity ' v ' or rotated with constant angular velocity ' ω ' in uniform and constant magnetic field ' B_0 ', which is same for all cases and extended upto infinity distance between points A and B directly in each case is ' l '. In case of pure rotation plane of rotation is perpendicular to direction of magnetic field. Potential difference between points B and A is $V_B - V_A$.



COLUMN-I

A) $V_B - V_A = B l v$

B) Rate of area swept by the conducting element $= \frac{dA}{dt} = v l$

C) Uniform electric field in the conductor

D) When velocity or angular velocity remains constant, then the external force required is zero

COLUMN-II

p) Case I

q) Case II

r) Case III

s) Case IV

Integer Type Questions

21. A conducting rod of length L is moving on a horizontal smooth surface. Magnetic field in the region is vertically downward and of magnitude B_0 . If centre of mass of the rod is translating with velocity v_0 and rod rotates about centre of mass with angular velocity v_0/L then potential difference between points O and A will be $\frac{x}{8} B_0 v_0 L$. Then x is



22. Two conducting rings P and Q of radii r and $2r$ rotate uniformly in opposite directions with centre of mass velocities $2v$ and v respectively on a conducting surface S. There is a uniform magnetic field of magnitude B perpendicular to the plane of the rings. The potential difference between the highest points of the two rings is $\frac{1}{8} Bvr$

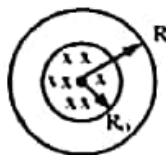
EXERCISE-III

(Induced Electric fields)

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The magnetic field (B) within a cylindrical region of radius R_0 increases linearly with time. A concentric circle with radius $R > R_0$ is drawn as shown in the figure. The electric field (E) acting tangential to the circle satisfies



- a) $E \propto R$ b) $E \propto \frac{1}{R}$ c) $E \propto \frac{1}{R^2}$ d) $E = \text{constant}$
2. Figure shows three regions of magnetic field, each of area A , and in each region magnitude of magnetic field decreases at a constant rate α . If $\vec{E}$ is induced electric field then value of line integral $\oint \vec{E} \cdot d\vec{l}$ along the given loop is equal to

- a) αA
b) $-\alpha A$
c) $3\alpha A$
d) $-3\alpha A$

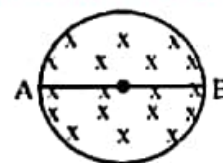


3. A uniform circular ring of radius R , mass m has uniformly distributed charge q . The ring is free to rotate about its own axis (which is vertical) without friction. In the space, a uniform magnetic field B , directed vertically downwards, exists in a cylindrical region. Cylindrical region of magnetic field is coaxial with the ring and has radius ' r ' greater than R . If magnetic field starts increasing at a constant rate $\frac{dB}{dt} = a$, angular acceleration of the ring will be

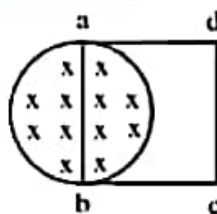
- a) $\frac{qR}{2m} \frac{dB}{dt}$ b) $\frac{q}{2m} \left(\frac{dB}{dt} \right)$ c) $\frac{qR}{m} \frac{dB}{dt}$ d) $\frac{qR}{m} \left(\frac{dB}{dt} \right)$

4. The radius of the circular conducting loop shown in figure is R . Magnetic field is decreasing at a constant rate α . Resistance per unit length of the loop is ρ . Then current in wire AB is : (AB is one of the diameters)

- a) $\frac{R\alpha}{2\rho}$ from A to B b) $\frac{R\alpha}{2\rho}$ from B to A
c) $\frac{2R\alpha}{\rho}$ from A to B d) zero



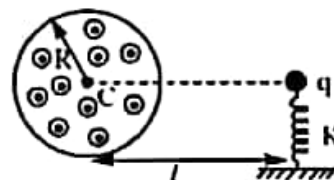
5. A uniform magnetic field B exists in a cylindrical region of radius 10 cm as shown in figure. A uniform wire of length 80 cm and resistance 4.0Ω is bent into a square frame and is placed with one side along a diameter of the cylindrical region. If the magnetic field increases at a constant rate of 0.010 T/s , find the current induced in the frame.



- a) $3.9 \times 10^{-5} \text{ A}$ b) $5 \times 10^{-5} \text{ A}$ c) $3.9 \times 10^{-5} \text{ A}$ d) $1 \times 10^{-6} \text{ A}$

6. There is a horizontal cylindrical uniform but time varying magnetic field increasing at a constant rate $\frac{dB}{dt}$ as shown. A charged particle having charge q and mass m is kept in equilibrium, at the top of a spring of spring constant K in such a way that it is on the horizontal line passing through the center of the magnetic field as shown in figure. The compression in the spring will be

- a) $\frac{1}{K} \left[mg - \frac{qR^2}{2l} \frac{dB}{dt} \right]$ b) $\frac{1}{K} \left[mg + \frac{qR^2}{l} \frac{dB}{dt} \right]$
c) $\frac{1}{K} \left[mg + \frac{2qR^2}{l} \frac{dB}{dt} \right]$ d) $\frac{1}{K} \left[mg + \frac{qR^2}{2l} \frac{dB}{dt} \right]$



7. A uniform magnetic field of intensity $B = C_1 t + C_2$ is directed into the plane of the paper exists in the cylindrical region of radius r . Where C_1 and C_2 are positive constant and t is time in sec. A loop in form of an equilateral triangle ABC of side length $2r$ is placed such that vertex C lies on axis of cylindrical region and plane of loop is perpendicular to axis. Resistance of unit length of wire of loop = $\lambda \Omega/\text{m}$. Find the potential difference $V_A - V_B$ at any $t = t$ sec

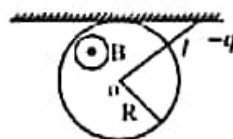
- a) $+\frac{C_1 \pi r^2}{6}$ b) $-\frac{C_1 \pi r^2}{6}$
c) $+\frac{C_1 \pi r^2}{9}$ d) $-\frac{C_1 \pi r^2}{9}$



8. A time varying magnetic field directed out of the plane of the paper exists in a cylindrical region of radius R whose centre is at O. The rate of change of magnetic field is constant and equal to $k(>0)$. A smooth rigid wall is tangential to this circular region. A charge $-q$ is placed in a gravity free

space nearest to the wall at a distance l from the centre of cylindrical region as shown in figure. The net force on the charge is ($R \ll l$) (Including the normal reaction from the wall)

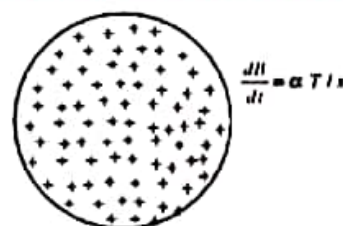
- a) zero
b) $\frac{qR^2 k}{2l}$
c) $\frac{qR^3 k}{2l^2}$
d) $\frac{qR^3 k}{l^2}$



More than One correct answer Type Questions

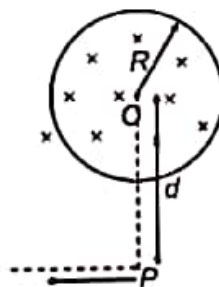
9. A time varying magnetic field is present in a cylindrical region of radius R as shown in figure (cross-sectional view). B is increasing with time; mark out the correct statement(s) for the given situation, r being the distance from centre of cylindrical region.

- a) For $r < R$, the induced electric field is proportional to r .
b) For $r > R$, the induced electric field is proportional to $\frac{1}{r}$.
c) For $r = R$, the induced electric field is maximum.
d) If a coaxial "non-conducting" ring of radius $\frac{R}{2}$ is placed in the magnetic field region, then emf induced in the ring is $\frac{\pi \alpha R^2}{4}$.



10. In a cylindrical region of radius R , there exists a time varying magnetic field B such that $\frac{dB}{dt} = K(>0)$. A charged particle having charge q is placed at the point P at a distance $d(>R)$ from its centre O . Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent up to infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s

- a) Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = R$
b) Work done by external agent is $\frac{q\pi R^2}{8} k$ if $d = 4R$
c) Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 4R$
d) Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 6R$



Linked Comprehension Type Questions

Passage - I :

A thin non-conducting ring of mass m , radius a , carrying a charge q can rotate freely about its own axis which is vertical. At the initial moment the ring was at rest and no magnetic field was present. At instant $t = 0$, a uniform magnetic field is switched on which is vertically downwards and increase with time according to the law $B = B_0 t$. Neglecting magnetism induced due to rotational motion of the ring. Now answer the following questions.

11. The angular acceleration of the ring and its direction of rotation as seen from above:

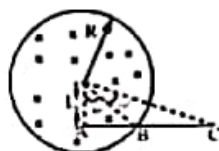
- a) $\frac{Eq}{2\pi m a}$ anti-clockwise b) $\frac{Eq}{m a}$ anti-clockwise c) $\frac{2Eq}{m a}$ clockwise d) $\frac{Eq}{m a}$ clockwise

12. The power developed by the forces acting on the ring, as a function of time:

- a) $\frac{E^2 q^2}{2m} t$ b) $\frac{E^2 q^2}{m} t$ c) $\frac{2E^2 q^2}{m} t$ d) $\frac{\sqrt{2}E^2 q^2}{m} t$

Passage - II :

A metallic rod is placed with a part of it in a cylindrical region of radius R where a uniform but time varying magnetic field exist which is varying in magnitude (B) with rate $\frac{dB}{dt} = \alpha (\alpha > 0)$, then



13. Induced potential difference between points A and B will be

- a) $\frac{L^2 \alpha}{2\sqrt{3}}$ b) $\frac{2L^2 \alpha}{\sqrt{3}}$ c) $\frac{\sqrt{3}L^2 \alpha}{2}$ d) zero

14. Potential difference between points B and C will be

- a) zero b) $\frac{\pi R^2 \alpha}{12}$ c) $\frac{\pi R L \alpha}{\sqrt{3}}$ d) $\frac{R^3 \alpha}{\sqrt{3} L \pi}$

15. Magnitude of induced electric field intensity at a point distant $2L$ from axis of cylindrical region of magnetic field will be

- a) $\frac{R^2 \alpha}{4L}$ b) $L\alpha$ c) $\frac{R\alpha}{2}$ d) $\frac{L^2 \alpha}{\sqrt{3}L}$

EXERCISE-IV

(Self induction, Mutual Induction & Magnetic Energy density)

LEVEL-I (MAIN)

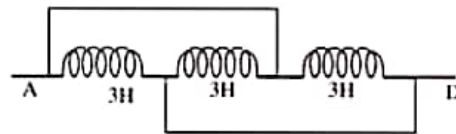
Straight Objective Type Questions

- An e.m.f of 5 volt is produced by a self inductance, when the current changes at a steady rate from 3A to 2A in 1 millisecond. The value of self inductance is
1) Zero 2) 5H 3) 5000H 4) 5 mH
- If a change in current of 0.01A in one coil produces a change in magnetic flux of 2×10^{-2} weber in the other coil, then the mutual inductance of the two coils in henry is
1) 0 2) 0.5 3) 2 4) 3
- A 50 mH coil carries a current of 2 ampere. The energy stored in joules is
1) 1 2) 0.1 3) 0.05 4) 0.5

Numerical Type Questions

- The number of turns of primary and secondary coils of a transformer are 5 and 10 respectively and the mutual inductance of the transformer is 25 henry. Now the number of turns in the primary and secondary of the transformers are made 10 and 5 respectively. The mutual inductance of the transformer (in henry) will be :

5. The inductance between A and D is in Henry.



6. Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A = 10 \text{ cm}^2$ and length $= 20 \text{ cm}$. If one of the solenoid has 300 turns and the other 400 turns, their mutual inductance (in H) is $X \times 10^{-4} \text{ H}$. The value of 'X' is
7. A current of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time in J/sec.

LEVEL-II (ADVANCED)

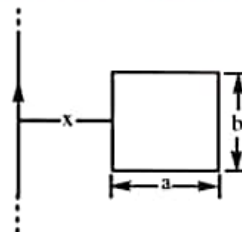
Straight Objective Type Questions

1. Mutual inductance of a pair of concentric conducting coils of radii a and b (b being very small and $a \gg b$) with their planes inclined at 30° is given by

a) $\frac{\sqrt{3}\mu_0\pi b^2}{4a}$ b) $\frac{\mu_0\pi b^2}{4a}$ c) $\frac{\sqrt{3}\mu_0 b^2}{2\pi a}$ d) $\frac{\sqrt{3}\mu_0\pi b^2}{8a}$

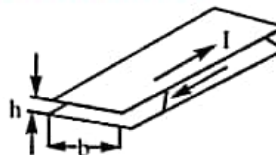
2. The mutual inductance of the system of long straight conductor and the rectangular loop is

a) $\frac{\mu_0}{2\pi} a \ln\left(\frac{a+x}{a}\right)$ b) $\frac{\mu_0}{2\pi} b \ln\left(\frac{x+a}{x}\right)$
 c) $\frac{\mu_0}{2\pi} b \ln\left(\frac{x+a}{a}\right)$ d) $\frac{\mu_0}{2\pi} a \ln\left(\frac{a+x}{x}\right)$



- *3. Calculate the inductance of a unit length of a double tape line if the tapes are separated by a distance h which is considerably less than their width b , namely, $b/h = 50$.

- a) 20 nH/m
 b) 25 nH/m
 c) 30 nH/m
 d) 15 nH/m



4. A superconducting round ring of radius a and inductance L was located in a uniform magnetic field of induction B . The ring plane was parallel to the vector B , and the current in the ring was equal to zero. Then the ring was turned through 90° so that its plane became perpendicular to the field. Find the current induced in the ring after the turn.

a) $I = 2\pi a^2 B/L$ b) $I = \pi a^2 B/2L$ c) $I = \pi a^2 B/L$ d) $I = 3\pi a^2 B/2L$

5. In the above problem find the work performed during the turn.

a) $W = \frac{1}{2}\pi^2 a^4 B^2 / L$ b) $W = \frac{1}{3}\pi^2 a^4 B^2 / L$ c) $W = \frac{1}{3}\pi^2 a^4 B^2 L$ d) $W = \frac{1}{2}\pi^2 a^4 B^2 L$

- *6. A long straight wire of circular cross-section is made of a non-magnetic material. The wire is of radius a . The wire carries a current I which is uniformly distributed over its cross-section. The energy stored per unit length in the magnetic field contained within the wire is

a) $\frac{U}{l} = \frac{\mu_0 I^2}{8\pi}$ b) $\frac{U}{l} = \frac{\mu_0 I^2}{16\pi}$ c) $\frac{U}{l} = \frac{\mu_0 I^2}{4\pi}$ d) $\frac{U}{l} = \frac{\mu_0 I^2}{2\pi}$

- *7. A long cylinder of radius a carrying a uniform surface charge rotates about its axis with an angular velocity ω . Find the magnetic field energy per unit length of the cylinder if the linear charge density equals λ and $\mu_r = 1$.

a) $W_l = \mu_0 \lambda^2 \omega^2 a^2 / 6\pi$ b) $W_l = \mu_0 \lambda^2 \omega^2 a^2 / 8\pi$
c) $W_l = \mu_0 \lambda \omega a / 6\pi$ d) $W_l = \mu_0 \lambda \omega a / 8\pi$

8. A thin uniformly charged ring of radius $a = 10$ cm rotates about its axis with an angular velocity $\omega = 100$ rad/s. Find the ratio of volume energy densities of magnetic and electric fields on the axis of the ring at a point moved from its centre by a distance $l = a$.

a) 2.1×10^{-15} b) 4.1×10^{-15} c) 6.1×10^{-15} d) 1.1×10^{-15}

Linked Comprehension Type Questions

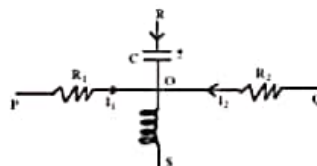
Passage :

$$R_1 = 2\Omega, R_2 = 3\Omega$$

$$C = 2F, L = 4H$$

$$i_1 = e^{-2t}, i_2 = 4A \text{ and}$$

$$V_c = 3e^{-2t}V$$



9. Current through inductor (i_L) is

a) $(2 - 4e^{-2t})$ A b) $(4 - 2e^{-2t})$ A c) $(11e^{-2t} - 4)$ A d) $4e^{-2t}$ A

10. Potential difference across inductor is (V_L)

a) $16e^{-2t}$ V b) $4e^{-2t}$ V c) $8e^{-2t}$ V d) $88e^{-2t}$ V

11. Potential difference between P and Q is (V_{PQ})

a) $(10e^{-2t} - 12)$ V b) $(12e^{-2t} - 10)$ V c) $(2e^{-2t} - 12)$ V d) $12(e^{-2t} - 1)$ V

12. Potential difference between P and R is

a) $7e^{-2t}$ V b) $-e^{-2t}$ V c) $5e^{-2t}$ V d) $10e^{-2t}$ V

EXERCISE-V

(L.R. Circuits)

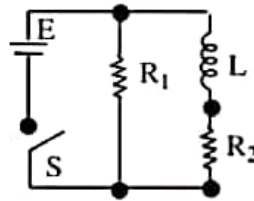
LEVEL-I (MAIN)

Straight Objective Type Questions

1. An ideal coil of 10 henry is joined in series with a resistance of 5 ohm and a battery of 5 volt. 2 second after joining, the current flowing in ampere in the circuit will be

1) e^{-1} 2) $(1 - e^{-1})$ 3) $(1 - e)$ 4) e

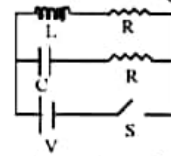
2. An inductor of inductance $L = 400 \text{ mH}$ and resistor of resistances $R_1 = 2\Omega$ and $R_2 = 2\Omega$ are connected to a battery of emf 12V as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at $t = 0$. The potential drop across L as a function of times is:



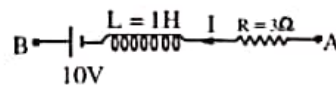
- 1) $\frac{12}{t} e^{-3t} \text{V}$ 2) $6(1 - e^{-t/0.2}) \text{V}$ 3) $12e^{-5t} \text{V}$ 4) $6e^{-5t} \text{V}$

Numerical Type Questions

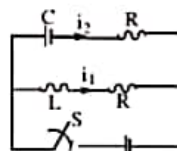
3. In a L - R decay circuit, the initial current is I . The total charge that has flown through the resistor till the energy in the inductor has reduced to one-fourth its initial value, is $\frac{LI}{XR}$. The value of ' X ' is
4. In the circuit shown in the figure, $R = \sqrt{\frac{L}{C}}$. Switch S is closed at time $t = 0$. The current through C and L would be equal after a time t equal to $CR \log(P)$. The value of ' P ' is



5. In the branch AB of a circuit, as shown in the figure, a current $I = (10t + 5)\text{A}$ is flowing, where t is the time in second. At $t = 0$, the value of $(V_A - V_B)$ is $(3x)$ volt. Find the value of ' x '.



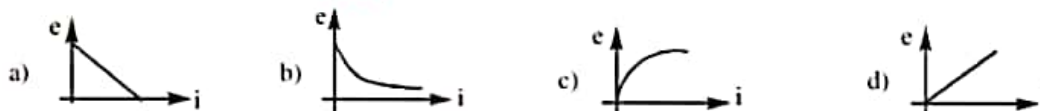
6. An LR circuit is connected to a battery at time $t = 0$. The energy stored in the inductor reaches half its maximum value at time $\frac{L}{R} \log[X]$. The value of ' X ' is
7. In an oscillating LC circuit the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic field is $\frac{Q}{\sqrt{N}}$. The value of ' N ' is
8. A pure inductor, pure capacitor and two pure equal resistors are connected as shown in the circuit, such that $L = 2R^2C$. At an instant $t = 0$ the switch is closed. If i_1 and i_2 are currents shown in the branches, at some instant of time $t = t_0$ the relation $\frac{di_1}{dt} + \frac{di_2}{dt} = 0$ is satisfied. Then $t_0 = RC(\ln \sqrt{k})$; where $k =$



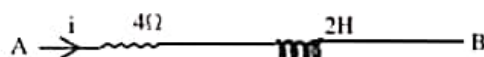
LEVEL-II (ADVANCED)

Straight Objective Type Questions

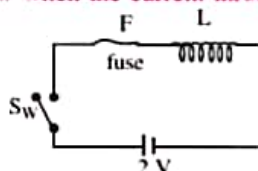
1. In an L-R circuit connected to a battery of constant e.m.f. E , switch S is closed at time $t = 0$. If e denotes the magnitude of induced e.m.f. across inductor and i the current in the circuit at any time t , then which of the following graphs shows the variation of e with i ?



2. In the given branch $i = 10e^{-4t}$ A. Then

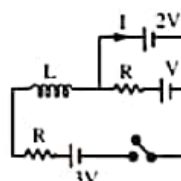


- a) $V_L = 80e^{-4t}$ and $V_A - V_B = -40e^{-4t}$
 b) $V_L = 80e^{-4t}$ and $V_A - V_B = 120e^{-4t}$
 c) $V_L = 0$ and $V_A - V_B = -40e^{-4t}$
 d) $V_L = 80e^{-4t}$ and $V_A - V_B = -80e^{-4t}$
3. For L-R circuit, the time constant is equal to
- a) twice the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 b) ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 c) half the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 d) square of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
4. In the circuit shown, the cell is ideal. The coil has an inductance of $4H$ and zero resistance. F is a fuse of zero resistance and will blow when the current through it reaches $5A$. The switch is closed at $t = 0$. The fuse will blow :

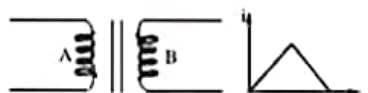


- a) just after $t = 0$
 b) after 2s
 c) after 5s
 d) after 10s
5. In the LR circuit shown, what is the variation of the current I as a function of time? The switch is closed at time $t = 0$.

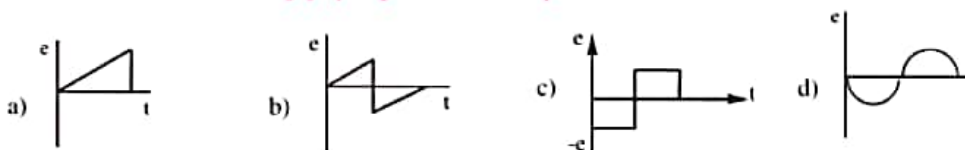
- a) $\frac{V}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$
 b) $\frac{V}{R} e^{-\frac{Rt}{L}}$
 c) $-\frac{V}{R} e^{-\frac{Rt}{L}}$
 d) None



6. A and B are two coils placed closely as shown. The current in coil A varies as shown in the graph.



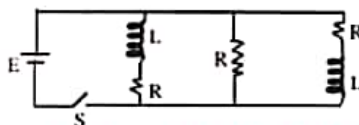
Then which of the following graphs gives the best representation of variation of induced emf in coil B ?



Linked Comprehension Type Questions

Passage - I :

The given circuit shows three identical resistors with resistance $R = 9 \Omega$, two identical inductors with inductance $L = 2 \text{ mH}$ and an ideal battery with emf $E = 18 \text{ V}$.



7. The current through the battery just after the switch is closed will be
a) 6A b) 4A c) 2A d) 1A
8. The current through the battery long time after the switch has been closed will be
a) 6A b) 4A c) 2A d) 1A
9. The current through the middle resistor long time after the switch has been closed will be
a) 6A b) 4A c) 2A d) 1A

Passage - II :

An LC circuit has capacitance C_1 and inductance L_1 . A second circuit has $C_2 = \frac{C_1}{2}$ and $L_2 = 2L_1$ and a third circuit has $C_3 = 2C_1$ and $L_3 = \frac{L_1}{2}$. Each is oscillatory circuit.

10. If in each case the capacitor was charged to same potential when connected to the inductor, maximum current will be generated in.
a) first circuit b) second circuit c) third circuit d) same in all circuit
11. Frequency of oscillation
a) is highest for first circuit b) is highest for second circuit
c) is highest for third circuit d) is same for all the three circuits
12. If I_1, I_2, I_3 are the maximum currents in those three circuits respectively
a) $I_1 > I_2 > I_3$ b) $I_3 > I_1 > I_2$ c) $I_3 > I_2 > I_1$ d) $I_1 < I_2 > I_3$

Matrix Matching Type Questions

13. Comparing L-C oscillations with the oscillations of spring-blocks system match the following table:

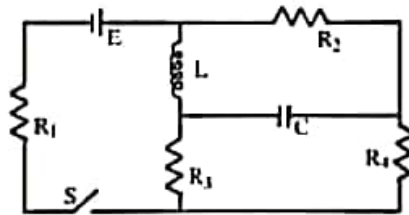
COLUMN - I
(LC oscillations)

- A) L
B) C
C) i
D) $\frac{di}{dt}$

COLUMN - II
(Spring-block oscillations)

- p) k
q) m
r) v
s) x
t) None

14. A circuit is shown in figure. At $t = 0$ a cell of emf 10 volt is connected and there is no initial charge on the capacitor and no initial current in any branch. Given $R_1 = R_2 = R_3 = R_4 = 1\Omega$, $C = 4\mu F$ and $L = 2\text{ mH}$. Match the column-I and column-II.



COLUMN - I

- A) $t = 0$ current in R_1 (in amp) is
 B) $t = \infty$ current in R_1 (in amp) is
 C) $t = \infty$ charge on capacitor (in μC) is
 D) $t = 0$ potential difference across R_4 (in volt)

COLUMN - II

- p) 2
 q) 4
 r) 6
 s) 8

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 1 2) 1 3) 4 4) 2 5) 1 6) 157.0 7) 0.0
 8) 5.0 9) 0.57 10) 2.0

LEVEL-II

- 1) c 2) c 3) b 4) b 5) c 6) b 7) a 8) a
 9) a,d 10) a 11) a 12) c 13) c 14) c 15) b 16) b
 17) A-qs; B-ps; C-ps; D-qs 18) A-r; B-p; C-s; D-q

EXERCISE-II

LEVEL-I

- 1) 1 2) 3 3) 4 4) 4 5) 2 6) 2 7) 0.02 8) 25.0
 9) 2.0 10) 6.25 11) 1.0

LEVEL-II

- 1) a 2) b 3) d 4) d 5) b 6) b 7) d 8) d
 9) b 10) c 11) b 12) a 13) b,d 14) b,c 15) a,d 16) a,b,c
 17) a 18) c 19) c 20) A-pqrs; B-pqr; C-qs; D-qs 21) 5
 22) 4

EXERCISE-III

LEVEL-II

- 1) b 2) b 3) b 4) d 5) a 6) d 7) d 8) c
 9) a,b,c,d 10) a,c,d 11) b 12) b 13) a 14) b 15) a

EXERCISE-IV

LEVEL-I

- 1) 4 2) 3 3) 2 4) 25.0 5) 1.0 6) 7.53 7) 16.0

LEVEL-II

- 1) a 2) b 3) b 4) c 5) a 6) b 7) b 8) d
 9) c 10) d 11) c 12) b

EXERCISE-V

LEVEL-I

1) 2 2) 3 3) 2.0 4) 2.0 5) 5.0 6) 3.41 7) 2.0 8) 16.0

LEVEL-II

1) a 2) a 3) a 4) d 5) c 6) c 7) c 8) a
 9) c 10) c 11) d 12) b 13) A-p; B-t; C-r; D-t
 14) A-q; B-r; C-s; D-p

PRACTICE SHEET

EXERCISE-I

(Magnetic flux, faraday's laws)

LEVEL-I (MAIN)

Straight Objective Type Questions

- Magnetic flux Φ (in weber) linked with a closed circuit of resistance 10 ohm varies with time t (in seconds) as $\Phi = 5t^2 - 4t + 1$. The induced electromotive force in the circuit at $t = 0.2 \text{ sec}$ is
 1) 0.4 volts 2) -0.4 volts 3) -2.0 volts 4) 2.0 volts
- A square coil of side 0.5 m has movable sides. It is placed such that its plane is perpendicular to uniform magnetic field of induction 0.2 T . If all the sides are allowed to move with a speed of 0.1 ms^{-1} for 4 sec outwards, average induced emf is
 1) Zero 2) 0.01 V 3) 0.028 V 4) 0.072 V
- A coil having n turns and resistance $R \Omega$ is connected with a galvanometer of resistance $4R \Omega$. This combination is moved in t second from field W_1 to W_2 weber. The induced current in circuit is :
 1) $\frac{n(W_2 - W_1)}{5Rt}$ 2) $\frac{W_2 - W_1}{5Rnt}$ 3) $\frac{n(W_2 - W_1)}{Rt}$ 4) $\frac{(W_2 - W_1)}{nRt}$
- A very small circular loop of area $5 \times 10^{-4} \text{ m}^2$ resistance 2Ω and negligible inductance is initially coplanar and concentric with a much larger fixed circular loop of radius 0.1 m . A constant current of 1 ampere is passed in larger loop and smaller loop is rotated with a angular velocity ' ω ' rads^{-1} about its diameter. The induced emf is $___ \times 10^{-9} \text{ V}$
 1) $\pi\omega^2 \sin \omega t$ 2) $\frac{\pi\omega}{2} \sin \omega t$ 3) $\frac{\pi\omega}{8} \cos \omega t$ 4) $\pi\omega \sin \omega t$

Numerical Type Questions

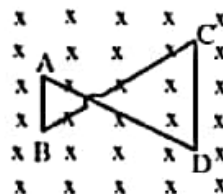
- A thin circular ring of area A is held perpendicular to an uniform magnetic field of induction B . A small cut is made in the ring and a galvanometer is connected across the ends such that the total resistance of the circuit is R . When the ring is suddenly squeezed to zero area, the charge flowing through the galvanometer is $\frac{BA}{NR}$. The value of ' N ' is
- A conducting metal circular-wire-loop of radius r is placed perpendicular to a magnetic field which varies with time as $B = B_0 e^{-t/\tau}$, where B_0 and τ are constant, at time $t = 0$. If the resistance of the loop is R then the heat generated in the loop after along time ($t \rightarrow \infty$) is $\frac{\pi^2 r^4 B_0^2}{2TR}$. The value of $(a+b)$ is

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are

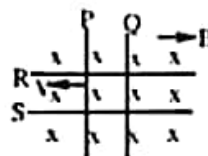
- a) B to A and D to C
b) A to B and C to D
c) A to B and D to C
d) B to A and C to D



2. The e.m.f. induced in a coil of wire, which is rotating in a magnetic field, does not depend on
a) the angular speed of rotation
b) the area of the coil
c) the number of turns in the coil
d) the resistance of the coil
3. Two identical coaxial circular loops carry a current i each circulating in the same direction. If the loops approach each other
a) the current in each will decrease
b) the current in each will increase
c) the current in each will remain the same
d) the current in one will increase and in other will decrease

4. Two identical conductors P and Q are placed on two frictionless rails R and S in a uniform magnetic field directed into the plane. If P is moved in the direction shown in figure with a constant speed then rod Q

- a) will be attracted towards P
b) will be repelled away from P
c) will remain stationary
d) may be repelled or attracted towards P



5. a conducting loop of radius R is present in a uniform magnetic field B perpendicular the plane of the ring. If radius R varies as a function of time ' t ', as $R = R_0 + t$. The e.m.f induced in the loop is

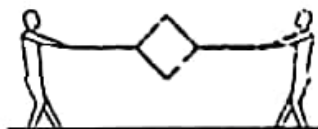
- a) $2\pi (R_0 + t)B$ clockwise
b) $\pi (R_0 + t)B$ clockwise
c) $2\pi (R_0 + t)B$ anticlockwise
d) zero



Linked Comprehension Type Questions

*Passage - I :

Figure shows a metallic square frame of edge a in a vertical plane. A uniform magnetic field B exists in the space in a direction perpendicular to the plane of the figure. Two boys pull the opposite corners of the square to deform it into a rhombus. They start pulling the corners at $t=0$ and displace the corners at a uniform speed u .

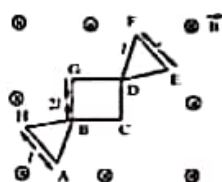


6. Find the induced emf in the frame at the instant when the angles at these corners reduce to 60° .
a) $5 B a v$
b) $2 B a v$
c) $1 B a v$
d) $1 B a v$

7. In the above problem, find the induced current in the frame at this instant if the total resistance of the frame is R .
 a) $3 B a v / R$ b) $9 B a v / R$ c) $2 B a v / R$ d) $10 B a v / R$
8. In the above problem, find the total charge which flows through a side of the frame by the time the square is deformed into a straight line.
 a) $a B / R$ b) $a^2 B^2 / R$ c) $2 a B / R$ d) $a^2 B / R$

Passage - II :

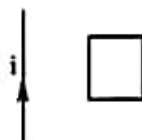
Diagram shows one square metallic loop and two equilateral triangular metallic loops touching square metallic loop at points B and D. All metallic loops are present in a region of uniform but time varying magnetic field, which is increasing in magnitude (B) at a rate $\frac{dB}{dt} = \alpha$ resistance of each side of triangular or square loop is r . Magnetic field is perpendicular to plane of each loop and points away from reader, then



9. Induced current in side BC is
 a) $\frac{l^2 \alpha}{r}$, along B to C b) $\frac{l^2 \alpha}{r}$, along C to B c) $\frac{\sqrt{3} l^2 \alpha}{4r}$, along B to C d) not possible to determine
10. Induced current in side AB is
 a) $\frac{l^2 \alpha}{3r}$, along A to B b) $\frac{l^2 \alpha}{8r}$, along A to B c) $\frac{l^2 \alpha}{4\sqrt{3}r}$, along A to B d) not possible to determine
11. Induced current in side DF is
 a) $\frac{l^2 \alpha}{\sqrt{2}r}$, along D to F b) $\frac{4l^2 \alpha}{\sqrt{3}r}$, along F to D c) $\frac{l^2 \alpha}{\sqrt{3}r}$, along D to F d) $\frac{l^2 \alpha}{4\sqrt{3}r}$ along F to D

Matrix Matching Type Questions

12. A square loop is placed near a long straight current carrying wire as shown. Match the following table.

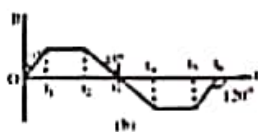
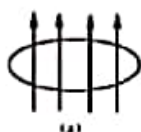
**COLUMN - I**

- A) If current is increased
 B) If current is decreased
 C) If loop is moved away from the wire
 D) If loop is moved towards the wire

COLUMN - II

- p) Induced current in loop is clockwise
 q) Induced current in loop is anticlockwise
 r) wire will attract the loop
 s) wire will repel the loop

13. A conducting loop is held in a magnetic field such that the field is oriented perpendicular to the area of the loop as shown in Fig-A. At  Observer (Positive direction of field)



any instant, magnetic flux density over the entire area has the same value but it varies with time as shown in Fig.B.

Match Column-I with Column-II

COLUMN - I

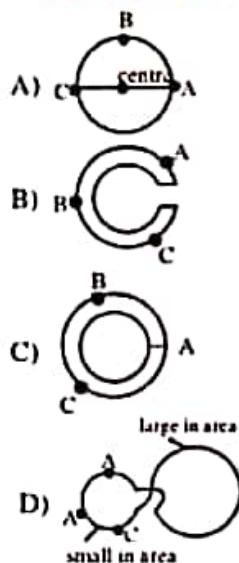
- A) Induced current in the coil is in the clockwise sense
- B) Induced current in the coil is in the anticlockwise sense
- C) Induced current is zero
- D) Induced current is maximum

COLUMN - II

- p) For $t_2 < t < t_3$
- q) For $t_3 < t < t_4$
- r) For $t_5 < t < t_6$
- s) For $t_4 < t < t_5$

14. Column-I shows thin plane conductors located in uniform magnetic field directed away from reader beyond the plane of paper. Column-II tells direction of resulting induced current in the conductors when magnetic field starts diminishing.

COLUMN - I



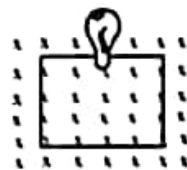
COLUMN - II

- p) Clockwise current in part ABC
- q) Anticlockwise current in part ABC
- r) No current in part ABC
- s) No current in some part of the conductor

Integer Type Questions

15. A coil having 100 turns and area of 0.001m^2 rotates freely about an axis passing through its diameter. The coil is placed perpendicular to a magnetic field of 1T . If the coil is rotated rapidly through an angle of 180° , and resistance of that coil is 10Ω , the charge will flow through the coil is $z \times 10^{-2}\text{C}$. Find z .
16. A direct current $I = 10\text{A}$ flows in a long straight round conductor. Find the magnetic flux through a half of wire's cross-section per one meter of its length is $_____\mu\text{wb/m}$

17. A square wire loop of 10.0 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt . The bulb will shine with full brightness if Δt is equal to $k \times 10\text{ms}$. Find the value of k



EXERCISE-II

(Motional emf)

LEVEL-I (MAIN)

Straight Objective Type Questions

1. A wire of length $2l$ is bent at mid point so that angle between the two halves is 60° . If it moves as shown with velocity ' v ' in a magnetic field ' B '. The induced emf will be
- Blv , P at high potential and Q at low potential
 - $2Blv$, P at high potential and Q at low potential
 - Blv , Q at high potential and P at low potential
 - $2Blv$, from Q at high potential and P at low potential



2. A player with 3 m long iron rod runs towards east with a speed of 30 km/hr. Horizontal component of earth's magnetic field is $4 \times 10^{-5} \text{ Wb/m}^2$. If he is running with rod in horizontal and vertical positions, then the potential difference induced between the two ends of the rod in two cases will be
- Zero in vertical position and $1 \times 10^{-3} \text{ V}$ in horizontal position
 - $1 \times 10^{-3} \text{ V}$ in vertical position and zero in horizontal position
 - Zero in both the cases
 - 1×10^{-3} in both the cases

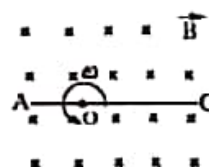
3. A conducting rod AC of length $4l$ is rotated about a point O in a uniform magnetic field $\vec{B}$ directed into the paper. $AO = l$ and $OC = 3l$. Then

1) $V_A - V_O = \frac{B\omega l^2}{2}$

2) $V_O - V_C = \frac{9}{2}B\omega l^2$

3) $V_A - V_C = B\omega l^2$

4) $V_A - V_C = \frac{9}{2}B\omega l^2$



4. The perfect formula used for calculating induced e.m.f in a rod moving in a uniform magnetic field is

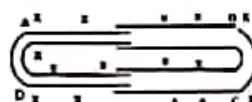
1) $\epsilon = \vec{B} \cdot (\vec{l} \times \vec{v})$

2) $\epsilon = \vec{B} \cdot (\vec{l} \cdot \vec{v})$

3) $\epsilon = \vec{B} \times (\vec{l} \cdot \vec{v})$

4) $\epsilon = \vec{B} \times (\vec{l} \times \vec{v})$

5. A conducting U-tube can slide inside another as shown in fig. maintaining electrical contacts between them. The magnetic field is perpendicular to plane of paper. If each tube move towards each other with constant speed v , then emf induced in the circuit in terms of B , l and v where l is the width of each tube.



1) Zero

2) $2B/v$

3) B/v

4) $-B/v$



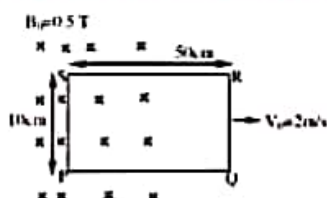
Numerical Type Questions

- A thin wire of length of 2m is perpendicular to the XY-plane. It is moved with velocity $\mathbf{v} = (2\hat{i} + 3\hat{j} + \hat{k})\text{ms}^{-1}$ through a region of magnetic induction $\mathbf{B} = (\hat{i} + 2\hat{j})\text{Wb/m}^2$. Then, potential difference induced between the ends of the wire in volts
- A bicycle is resting on its stand in east-west direction and rear wheel is rotating at 100rpm. length of each spoke is 30cm. and vertical component of Earth's magnetism is $1.5 \times 10^{-5}\text{T}$. If the emf induced in the spokes is $3\pi \times 10^{-6}\text{V}$, then the angle of dip at that place is $\tan^{-1}\left[\frac{a}{b}\right]$. The value of (a+b) is

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- A rectangular loop PQRS (10cm $\times$ 50 cm) is placed in a plane perpendicular to uniform magnetic induction field $B_0 = 0.5\text{T}$ and pulled out of the field region with a constant velocity $V_0 = 2\text{m/s}$. At a certain instant shown in the figure QR is out side the field region.



The resistance per unit length of the wire is $1\ \Omega/\text{m}$. $V_P - V_Q$ will be

- $\frac{11}{120}\text{V}$
 - $\frac{-11}{120}\text{V}$
 - $\frac{1}{120}\text{V}$
 - $\frac{-1}{120}\text{V}$
- A rod of length 10cm made up of conducting and non-conducting material (shaded part is non-conducting). The rod is rotated with constant angular velocity 10rad/s about point O, in constant magnetic field of 2T as shown in the figure. The induced emf between the points A and B of rod will be :



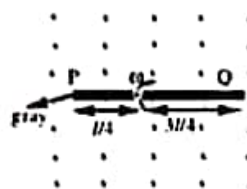
- 0.029V
 - 0.1V
 - 0.051V
 - 0.064V
- A conducting rod of length l is rotating with constant angular velocity ω about an axis OO' in a uniform magnetic field B as shown in figure. The emf induced between ends P and Q will be

a) $\frac{1}{4}B\omega l^2$

b) $\frac{5}{10}B\omega l^2$

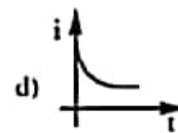
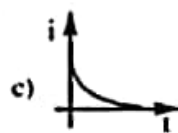
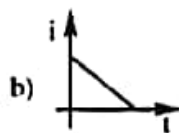
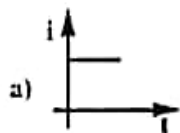
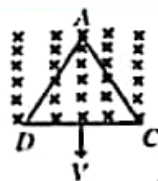
c) Zero

d) $\frac{1}{2}B\omega l^2$



- An equilateral triangular loop ADC of some finite magnetic field $\vec{B}$ as shown in the figure. At time $t = 0$, side DC of loop is at edge of the magnetic field. Magnetic field is perpendicular to the paper inwards (or perpendicular to the plane of the coil). The induced current versus time graph will be as





5. The magnetic field in a region is given by $\vec{B} = B_0 \left(1 + \frac{x}{a} \right) \hat{k}$. A square loop of edge - length d is placed with its edge along x & y axis. The loop is moved with constant velocity $\vec{V} = V_0 \hat{i}$. The emf induced in the loop at that instant is

a) $\frac{V_0 B_0 d^2}{a}$

b) $\frac{V_0 B_0 d^2}{2a}$

c) $\frac{V_0 B_0 a^2}{d}$

d) None

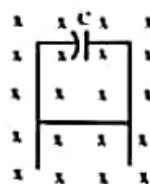
6. A wire of mass m and length l can slide freely on a pair of smooth, vertical rails. A magnetic field B exists in region in the direction perpendicular to the plane of the rails. The rails are connected at the top end by a capacitor of capacitance C . Find the acceleration of the wire neglecting any electric resistance.

a) $\frac{mg}{m + CB^2 l^2}$

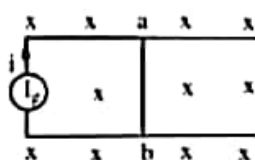
b) $\frac{mg}{m - CB^2 l^2}$

c) $\frac{2mg}{m - CB^2 l^2}$

d) $\frac{5mg}{m + CB^2 l^2}$



7. The current generator I_g shown in figure, sends a constant current i through the circuit. The wire ab has a length l and mass m and can slide on the smooth, horizontal rails connected to I_g . The entire system lies in a vertical magnetic field B . Find the velocity of the wire as a function of time.



a) $il Bt/m$ away from the generator

b) $2il Bt/m$ away from the generator

c) $2il Bt/m$ towards the generator

d) $il Bt/2m$ towards the generator

More than One correct answer Type Questions

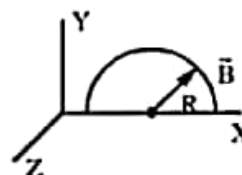
8. A semicircular conducting ring of radius R is placed in xy plane, as shown in the figure. A uniform magnetic field is set up along the x -axis. No emf will be induced in the ring, if

a) it moves along the x -axis

b) it moves along the y -axis

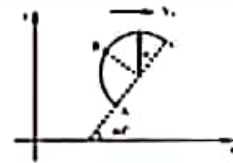
c) it moves along the z -axis

d) it remains stationary



9. A semicircular conducting metallic loop of radius R is moving in x - y plane in a region of uniform magnetic field given by $\vec{B} = B_0(\hat{i} + 2\hat{j} + 3\hat{k})$ with velocity $\vec{V} = V_0\hat{i}$ as shown in diagram. Then

- induced emf across its ends will be $3\sqrt{3}V_0B_0R$
- potential of end C is less than potential of end A
- point B and point C will be at same potential
- we can get two points on loop which will be at same potential

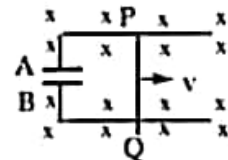


10. A conducting loop rotates with constant angular velocity about its fixed diameter in a uniform magnetic field. Whose direction is perpendicular to that fixed diameter.

- The emf will be maximum at the moment when flux is zero.
- The emf will be '0' at the moment when flux is maximum.
- The emf will be maximum at the moment when plane of the loop is parallel to the magnetic field
- The phase difference between the flux and the emf is $\pi/2$

11. A conducting rod PQ of length $L = 1.0$ m is moving with a uniform speed $v = 20$ m/s in a uniform magnetic field $B = 4.0$ T directed into the paper. A capacitor of capacity $C = 10$ mF is connected as shown in figure. Then

- $q_A = +800$ mC and $q_B = -800$ mC
- $q_A = -800$ mC and $q_B = +800$ mC
- $q_A = 0 = q_B$
- charged stored in the capacitor increases exponentially with time

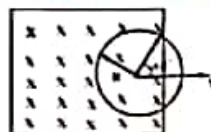


12. A metallic ring of radius R is moving in a region of uniform magnetic field of magnitude B_0 with speed V as shown. then

- Potential difference between points A and B is $\frac{B_0RV}{2}$
- Potential difference between points A and B is $\frac{B_0RV}{4}$
- Potential difference between points C and D is B_0RV
- Potential difference between points C and E is zero



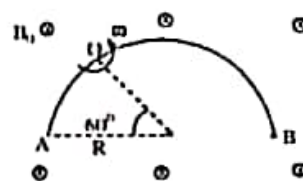
13. A circular conducting loop of radius R and resistance per unit length λ is pulled out from a region of uniform magnetic field with constant velocity v . The situation shown in figure is at $t = 0$. Mark out the correct statement(s)



- Just after $t=0$, i.e., the motion starts, the induced current in the loop is $\frac{\sqrt{3}Bv}{2\pi\lambda}$
- Current will be induced in the loop for total time interval of $\frac{3R}{2v}$
- Maximum induced current in loop is $\frac{Bv}{\pi\lambda}$ at $t = \frac{R}{2v}$
- Induced current is in clockwise direction for $t = 0$ to $\frac{R}{2v}$ and there after it becomes in anti-clockwise direction

14. A semi circular rigid metallic wire, of radius R is rotating with constant angular speed ω about an axis passing through point O and perpendicular to the plane of the semicircular wire in a region of uniform magnetic field ' B_0 ' as shown then

- a) Potential difference between points O and A is $\frac{B_0 R^2 \omega}{4}$
 b) Potential difference between points O and B is $\frac{3B_0 R^2 \omega}{2}$
 c) Electric Potential of point A is more than point B .
 d) Potential difference between points A and B is $B_0 R^2 \omega$



Linked Comprehension Type Questions

Passage - I :

A conducting ring of radius a is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field B which exists everywhere. The rotational velocity is ω .



15. Choose the correct statement(s) related to the potential of the points P , Q and R
- a) $V_P - V_O > 0$ and $V_R - V_O < 0$ b) $V_P = V_R > V_O$
 c) $V_O > V_P = V_Q$ d) $V_Q - V_P = V_P - V_O$
16. Choose the correct statement(s) related to the magnitude of potential differences
- a) $V_P - V_O = \frac{1}{2} B \omega a^2$ b) $V_P - V_Q = \frac{1}{2} B \omega a^2$ c) $V_Q - V_O = 2 B \omega a^2$ d) $V_P - V_R = 2 B \omega a^2$
17. Choose the correct statement(s) related to the induced current in the ring
- a) Current flows from Q to P to O to R to Q
 b) Current flows from Q to R to O to R to Q
 c) Current flows from Q to P to O and from Q to R to O
 d) No current flows

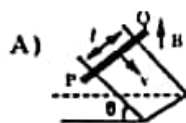
Passage - II :

A train of mass 100 tons ($1 \text{ ton} = 1000 \text{ kg}$) runs on a meter gauge track (distance between the two rails is 1 m .) The coefficient of friction between the rails and the train is 0.045 . The train is powered by an electric engine of 90% efficiency. The train is moving with uniform speed of 72 Kmph at its highest speed limit. Horizontal and vertical component of earth's magnetic field are $B_H = 10^{-5} \text{ T}$ and $B_V = 2 \times 10^{-5} \text{ T}$. Assume the body of the train and rails to be perfectly conducting.

18. The electrical power consumed by the train is
- a) 1.11 MW b) 1 MW c) 0.50 MW d) 0.90 MW
19. The potential difference between the two rails is
- a) 0.1 mV b) 0.2 mV c) 0.4 mV d) 0.8 mV
20. If now a resistor of 10^{-3} W is attached of between the two rails, the extra units of energy (electricity) consumed during a 324 km run of the train is (1 unit of power = 1 kW hour) (assume the speed of train to remain unchanged)
- a) $8 \times 10^{-4} \text{ KW hour}$ b) $8 \times 10^{-5} \text{ KW hour}$ c) $8 \times 10^{-6} \text{ KW hour}$ d) $8 \times 10^{-7} \text{ KW hour}$

Matrix Matching Type Questions

21. COLUMN - I



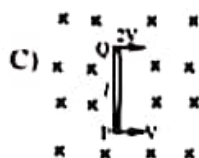
A)

A conducting rod $PQ = l$ slides with constant velocity v down two parallel rails separated by distance ' l ' and inclined at angle $\theta = 60^\circ$ to the horizontal. A uniform magnetic induction field B exists in vertically upward direction.



B)

A conducting ring with a conducting spoke $PQ = l$ rolls without sliding, the centre of the ring moves with constant velocity v . A uniform magnetic induction field B exists perpendicular to the plane of the ring as shown. At the position shown.



C)

A conducting rod $PQ = l$ moves in a plane perpendicular to uniform magnetic induction B with the velocities of the two ends as shown. At the position shown.



D)

A magnetic induction varying with time T as $\frac{dB}{dT} = \alpha$ (constant) exists in a cylindrical region. A uniform conducting ring is placed inside the region as shown.

COLUMN - II

p) $V_P - V_Q = vBl/2$

q) The magnitude of the electric field inside the rod/spoke/ring as we go from P to Q is constant

r) The magnitude of the electric field inside the rod/spoke/ring as go from P to Q increases.

s) The magnitude of the electric field inside the rod/spoke/ring is zero.

Integer Type Questions

22. A circular loop of radius 1m is kept in a magnetic field of strength 2T (plane of loop is perpendicular to the direction of magnetic field). Resistance of the loop wire is $\frac{2}{\pi}\Omega/\text{m}$. A conductor of length 2m is sliding with a speed 1m/s as shown in the figure. Find the instantaneous force acting on the rod: [assume rod has negligible resistance] $___\times 2\text{N}$



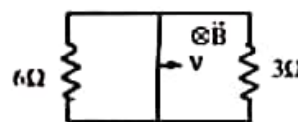
23. A rod of length 2m has translational velocity of 0.5 ms^{-1} in a direction making an angle of 30° with its length. The plane along which the rod is moving is normal to the magnetic field of induction 2T . The emf induced between the ends of the rod is

24. A conducting rod of length L is moving on a horizontal smooth surface. Magnetic field in the region is vertically downward and of magnitude B_0 . If centre of mass of the rod is translating with velocity v_0 and rod rotates about centre of mass with angular velocity v_0/L .



then potential difference between points O and A will be $\frac{x}{8} B_0 v_0 L$. Then x is

25. A rectangular loop with a sliding connector of length $l = 1.0\text{m}$ is situated in a uniform magnetic field $B = 2\text{T}$ perpendicular to the plane of loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in figure. The external force required to keep the connector moving with a constant velocity $v = 2\text{ m/s}$ is



- *26. A metallic wire is initially at rest along the curve $y = \frac{\lambda}{\mu} \sin\left(\frac{\pi x}{\lambda}\right)$, $x \in (0, 2\lambda)$. It is moved along the x - y plane parallel to the line $x = y$, with a speed u . There is a uniform magnetic field $2B_0 \hat{k}$ in the region. The maximum magnitude of induced emf between any two points on the wire is $\sqrt{\alpha} B_0 \lambda u$. The value of α is

EXERCISE-III

(Induced Electric field)

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. A uniform magnetic field of induction B is confined to a cylindrical region of radius R . The magnetic field is increasing at a constant rate of dB/dt (tesla/second). An electron of charge q , placed at the point P on the periphery of the field experiences an acceleration :

- a) $\frac{1}{2} \frac{eR}{m} \frac{dB}{dt}$ towards left b) $\frac{1}{2} \frac{eR}{m} \frac{dB}{dt}$ towards right
c) $\frac{eR}{m} \frac{dB}{dt}$ towards left d) zero.



2. A non conducting ring having charge q uniformly distributed over its circumference is placed on a rough horizontal surface. A vertical time varying magnetic field $B = 4t^2$ is switched on at time $t = 0$. Mass of the ring is m and radius is R , the ring starts rotating after 2 seconds. The coefficient of friction between the ring and the table is

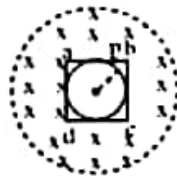
- a) $\frac{4qR}{mg}$ b) $\frac{2qR}{mg}$ c) $\frac{8qR}{mg}$ d) $\frac{qR}{2mg}$

3. A triangular wire frame (each side = 2m) is placed in a region of time variant magnetic field having $dB/dt = \sqrt{3}\text{ T/s}$. The magnetic field is perpendicular to the plane of the triangle. The base of the triangle AB has a resistance 1Ω while the other two sides have resistance 2Ω each. The magnitude of potential difference between the points A and B will be

- a) 0.4 V b) 0.6 V c) 1.2 V d) None



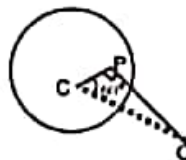
4. A uniform magnetic field B exists in a cylindrical region, shown dotted in figure. The magnetic field increases at a constant rate dB/dt . Consider a circle of radius r coaxial with the cylindrical region. Find the magnitude of the electric field E at a point on the circumference of the circle.



- a) $\frac{r}{3} \frac{dB}{dt}$ b) $\frac{r}{2} \frac{dB}{dt}$ c) $\frac{dB}{dt}$ d) $\frac{r}{2}$

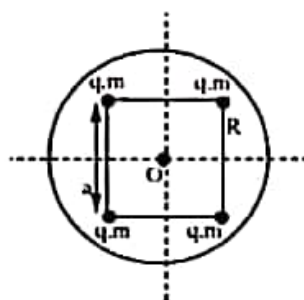
- *5. A magnetic field, confined in a cylindrical region of radius R , is changing at the rate of 4Ts^{-1} . A conducting rod PQ of length $\sqrt{\frac{3}{2}}R$ is placed in the region as shown. The induced emf across the rod will be

- a) $\frac{R^2}{24}(\pi+6)$ b) $\frac{\pi R^2}{24}$
c) $\frac{\pi R^2}{6}$ d) $\frac{R^2}{6}(\pi+6)$



More than One correct answer Type Questions

6. Four identical charged particles each of mass 0.1 kg and charge 2C connected to each other via massless non-conducting rods of equal length. The whole arrangement is placed in a cylindrical region carrying a uniform magnetic field as shown in the figure ($B_0=4\text{T}$, $a=1\text{m}$). Suddenly the magnetic field is switched off. Then choose the correct statement(s)



- a) The angular momentum of the system is 8 Nm/s
b) The angular momentum of the system is $8\sqrt{2}\text{ Nm/s}$
c) Angular velocity of the system is 40 rad/s
d) The magnetic field at point O immediately after the magnetic field is switched off is non zero

7. In the figure shown there exists a uniform time varying magnetic field $B = [(4 \text{ T/s}) t + 0.3 \text{ T}]$ in a cylindrical region of radius 4 m. An equilateral triangular conducting loop is placed in the magnetic field with its centroid on the axis of the field and its plane perpendicular to the field.

- a) e.m.f. induced in any one rod is 16 V
 b) e.m.f. induced in the complete ΔABC is $48\sqrt{3}$ V
 c) e.m.f. induced in the complete ΔABC is 48 V
 d) e.m.f. induced in any one rod is $16\sqrt{3}$ V



Linked Comprehension Type Questions

*Passage :

A very long solid conducting cylinder of radius R is placed in a constant and uniform magnetic field B which is parallel to the axis of the cylinder. Now the cylinder is rotated with a constant angular velocity ω as shown in the figure. Charge and mass of electron can be taken as e and m respectively



8. The electric field inside the cylinder as a function of radial distance r is

- a) $\left(\omega B - \frac{m\omega^2}{e}\right)r$ b) $E(r) = \frac{m\omega^2}{e}r$ c) $E(r) = \omega Br$ d) $E(r) = \left(\omega B + \frac{m\omega^2}{e}\right)r$

9. The potential difference between the axis of the cylinder and circumference is

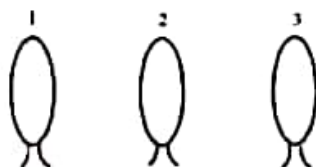
- a) $\Delta V = \left(\omega B - \frac{m\omega^2}{e}\right)\frac{R^2}{2}$ b) $\Delta V = \frac{m\omega^2 R^2}{2}$ c) $\Delta V = \frac{\omega B R^2}{2}$ d) $\Delta V = \left(\omega B + \frac{m\omega^2}{e}\right)\frac{R^2}{2}$

10. The volume charge density of the cylinder is

- a) $\omega B \epsilon_0$ b) $\frac{m\omega^2}{e} \epsilon_0$ c) $2\omega \epsilon_0 \left(B - \frac{m\omega}{e}\right)$ d) $\omega \epsilon_0 \left(B + \frac{m\omega}{e}\right)$

Matrix Matching Type Questions

11. Three coils are placed in front of each other as shown. Currents in 1 and 2 are in same direction, while that in 3 is in opposite direction. Match the following table :



COLUMN - I

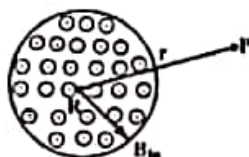
- A) When current in 1 is increased
 B) When current in 2 is increased
 C) When current in 3 is increased

COLUMN - II

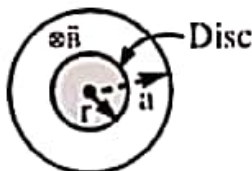
- p) current in 1 will increase
 q) current in 2 will increase
 r) current in 3 will increase
 s) None

Integer Type Questions

12. A long solenoid of cross-sectional radius $a = 10$ cm has a thin insulated wire ring tightly put on its winding. One half of the ring has the resistance η -times that of the other half. The magnetic induction produced by the solenoid varies with time as $B = bt$ where $b = 4\text{T/s}$ (constant). Now the magnitude value of electric field strength in the ring is found to be $10 \times 10^{-2} \text{ N/C}$. Find the value of η .
13. For the situation shown in figure, the magnetic field changes according to the expression $B = (2t^3 - 4t^2 + 0.8\text{T})$ and $r = 2R = 5\text{cm}$. If the force on electron at P is zero at $t = x/6\text{sec}$, then, x is



14. A uniform disc of radius r and mass m is charged uniformly with the charge q . This disc is placed flat on a rough horizontal surface having coefficient of friction μ . A uniform magnetic field is present in a circular region ($a > r$) but varying as kt^3 as shown in figure. Find the time in second after which the disc begins to rotate. (Given $r = 1\text{ m}$, $m = 18\text{ kg}$, $q = 1\text{C}$, $\mu = 0.1$, $K = 4$, $g = 10\text{ m/s}^2$)

**EXERCISE-IV***(Self Induction, Mutual Induction & Magnetic Energy density)***LEVEL-I (MAIN)**Straight Objective Type Questions

1. A small square loop of wire of each side l is kept inside a large square loop of wire of each side ' L ' ($L \gg l$). The loops are coplanar and their centres coincide. The mutual inductance of the system is proportional to (current is in the outer loop)
- 1) l/L 2) $\frac{l^2}{L}$ 3) L/l 4) $\frac{L^2}{l}$
2. When current breaks in primary coil current reaches to zero in 10^{-4} second. Emf induced in the secondary coil is $20,000\text{V}$ and mutual inductance between the coils is 5H . The maximum current in the primary before the break is
- 1) 0.2 amp 2) 0.4 amp 3) 4 amp 4) 2 amp
3. Two different coils have self - inductances $L_1 = 8\text{ mH}$ and $L_2 = 2\text{ mH}$. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time, the current, the induced voltage and the energy stored in the first coil are i_1 , V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are i_2 , V_2 and W_2 respectively. Then :
- a) $\frac{i_1}{i_2} = \frac{1}{4}$ b) $\frac{i_1}{i_2} = 4$ c) $\frac{W_1}{W_2} = \frac{1}{4}$ d) $\frac{V_1}{V_2} = 4$
- 1) a, d are true 2) c, d are true 3) a, c, d are true 4) Only b is true

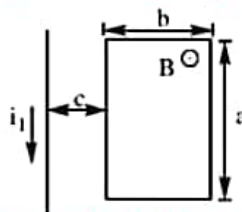
Numerical Type Questions

4. The coefficient of mutual inductance of two circuits A and B is 3mH and their respective resistance are 10 Ω and 4 Ω . The value of current (in A) should change in 0.02s in circuit A, so that to get current in B should be 0.006A is
5. The length of wire required to manufacture a solenoid of length X and self inductance L is $\sqrt{\frac{XL}{\mu_0}}$. The value of 'X' is

LEVEL-II (ADVANCED)

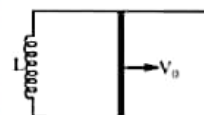
Straight Objective Type Questions

1. A long straight wire is placed along the axis of a circular ring of radius R . The mutual inductance of this system is
- a) $\frac{\mu_0 R}{2}$ b) $\frac{\mu_0 \pi R}{2}$ c) $\frac{\mu_0}{2}$ d) 0
2. Two long parallel wires whose centres are at distance d apart carry equal currents in opposite directions. If the flux within wires is neglected, the inductance of such arrangement of wire of length l and radius a will be
- a) $L = \frac{\mu_0 l}{2\pi} \log_e \frac{d-a}{a}$ b) $L = \frac{\mu_0 l}{\pi} \log_e \frac{d}{a}$ c) $L = \frac{\mu_0 l}{2\pi} \log_e \frac{a}{d}$ d) none
3. The mutual inductance between the rectangular loop and the long straight wire as shown in figure is M .
- a) $M = \text{Zero}$ b) $M = \frac{\mu_0 a}{4\pi} \ln \left(1 + \frac{c}{b} \right)$
- c) $M = \frac{\mu_0 b}{2\pi} \ln \left(\frac{a+c}{b} \right)$ d) $M = \frac{\mu_0 a}{2\pi} \ln \left(1 + \frac{b}{c} \right)$
4. Two coil A and B have coefficient of mutual inductance $M = 2\text{H}$. The magnetic flux passing through coil A changes by 4 weber in 10 seconds due to the change in current in B. Then
- a) change in current in B in this time interval is 0.5 A
- b) the change in current in B in this time interval is 2A
- c) the change in current in B in this time interval is 8A
- d) a change in current of 1A in coil A will produce a change in flux passing through B by 4 weber



More than One correct answer Type Questions

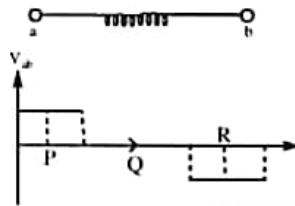
- *5. A π shaped conductor containing an inductance L is placed with its plane perpendicular to a uniform magnetic field B . A conducting rod of mass m can slide freely on this conductor as shown in figure. At $t = 0$, the rod is imparted an initial velocity V_0 . Then select correct option (s)



- a) time period of oscillation of rod is $2\pi\sqrt{\frac{Lm}{B^2 l^2}}$
- b) current (i) in the circuit as a function of time (t) is given by $i = \frac{m\omega V_0}{Bl} \sin \omega t$ (where $\omega = \sqrt{\frac{B^2 l^2}{Lm}}$)
- c) maximum acceleration of rod is $\sqrt{\frac{B^2 l^2 V_0^2}{Lm}}$
- d) rod will perform simple harmonic motion.

Matrix Matching Type Questions

6. In the figure V_{ab} versus time graph along an inductor is shown. Match the following.

**COLUMN - I**

- A) At P, if current is from b to a it must be
 B) At Q, if current is from a to b it must be
 C) At R, if current is from a to b it must be

COLUMN - II

- p) increasing
 q) decreasing
 r) constant

Integer Type Questions

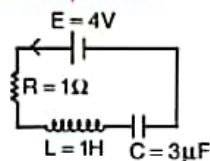
7. Two coils, 1 and 2 have a mutual inductance $M = 5\text{H}$ and resistance $R = 10\Omega$ each. A current flows in coil 1, which varies with time as: $I_1 = 2t^2$, where t is time. Find the total charge (in C) that has flown through coil 2 between $t = 0$ and $t = 2\text{s}$.

EXERCISE-V

(L.R. Circuits & L.C. Oscillations)

LEVEL-I (MAIN)Straight Objective Type Questions

- The time constant of an inductance coil is $3 \times 10^{-3}\text{ s}$. When a 90Ω resistance is joined in series, the time constant becomes $0.5 \times 10^{-3}\text{ s}$. The inductance and resistance of the coil are
 1) 54 mH; 18Ω 2) 14 mH; 42Ω 3) 42 mH; 14Ω 4) 14 mH; 60Ω
- An emf of 15 volt is applied in a circuit containing 5 henry inductance and 10 ohm resistance. The ratio of the currents in time $t = \infty$ and at $t = 1$ second is :
 1) $\frac{e^{1/2}}{e^{1/2}-1}$ 2) $\frac{e^2}{e^2-1}$ 3) $1 - e^{-1}$ 4) e^{-1}
- The current in the given circuit is increasing with a rate 4 amp/s. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be

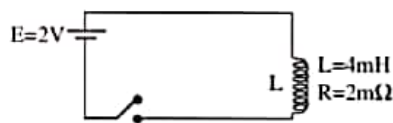


- 1) $4\mu\text{C}$ 2) $5\mu\text{C}$ 3) $6\mu\text{C}$ 4) $3\mu\text{C}$
- A coil of resistance 10 ohm and inductance 5 henry is connected to a 100 volt battery. Then, the energy stored in the coil. In the form of magnetic field (In steady state)
 1) 250 J 2) 1000 J 3) 125 J 4) 750 J

5. Coil of inductance 300 mH and resistance 2Ω is connected to a source of voltage 2V. The current reaches half of its steady state value in
 1) 0.05 S 2) 0.1 S 3) 0.15 S 4) 0.3 S
6. Two coils of self-inductance 2 mH and 8 mH are placed so close together that the effective flux in one coil is completely linked with the other. The mutual inductance between these coils is
 1) 16 mH 2) 10 mH 3) 6 mH 4) 4 mH

Numerical Type Questions

7. A resistance with an inductor $8 \times 10^{-2} \text{H}$ has same time constant as it has with a capacitor of $2\mu\text{F}$. The value of resistance in ohm is
8. The cell in the circuit shown in figure is ideal. The coil has an inductance of 4mH and a resistance of $2\text{m}\Omega$. The switch is closed at $t = 0$. The amount of energy stored in the inductor at $t = 2\text{s}$ is (take $e = 3$) in Joules.

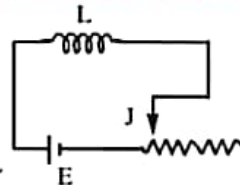


LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. In the circuit shown, the jockey J is pulled towards right so that the resistance in the circuit is increasing. Its value at some instant is R. Then current in the circuit at this instant will be

- a) E/R b) $< (E/R)$
 c) $> (E/R)$
 d) $< (E/R)$ or $> (E/R)$ depending on the value of L

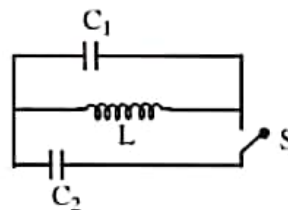


2. An LR circuit with a battery is connected at $t = 0$. Which of the following quantities is not zero just after the connection

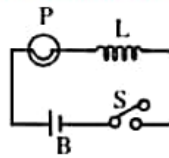
- a) current in the circuit b) magnetic field energy in the inductor
 c) power delivered by the battery d) emf induced in the inductor

3. At a moment ($t = 0$) when charge on capacitor C_1 is zero, the switch is closed. If I_0 be the current through inductor at that instant, for $t > 0$, [Initial charge on C_2 is also zero]

- a) maximum current through inductor equals $I_0/2$.
 b) maximum current through inductor equals $\frac{C_1 I_0}{C_1 + C_2}$
 c) maximum charge on $C_1 = \frac{C_1 I_0 \sqrt{LC_1}}{C_1 + C_2}$
 d) maximum charge on $C_1 = I_0 C_1 \sqrt{\frac{L}{C_1 + C_2}}$

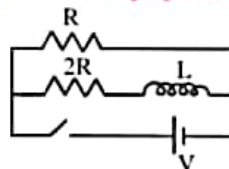


4. In figure, a lamp P is in series with an iron-core inductor L. When the switch S is closed, the brightness of the lamp rises relatively slowly to its full brightness than it would do without the inductor. This is due to

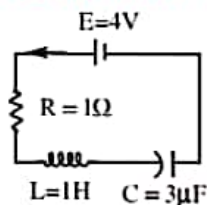


- a) the low resistance of P
b) the induced-emf in L
c) the low resistance of L
d) the high voltage of the battery B
5. A capacitor of capacitance 2 mF is charged to a potential difference of 12 V. It is then connected across an inductor of inductance 0.6 mH. The current in the circuit when the potential difference across the capacitor is 6 V is ;
- a) 3.6 A
b) 2.4 A
c) 19 A
d) 0.6 A
6. An induction coil stores 32 joules of magnetic energy and dissipates energy as heat at the rate of 320 watts, when a steady current of 4 amperes is passed through it. Find the time constant of the circuit when the coil is joined across a battery.
- a) 0.2 s
b) 0.1 s
c) 0.3 s
d) 0.4 s

7. The ratio of time constant in charging and discharging in the circuit shown in figure is

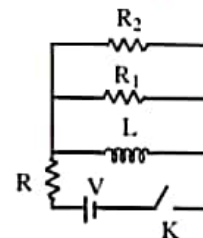


- a) 1 : 1
b) 3 : 2
c) 2 : 3
d) 1 : 3
8. The current in the given circuit is increasing with a rate of 4 amp/s. The charge on the capacitor at an instant when the current in the circuit is 2 amp will be ;



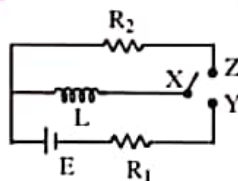
- a) 4 μC
b) 5 μC
c) 6 μC
d) none of these
9. In the circuit shown, initially switch k is closed for long time. At time $t = 0$, switch k is open. Then total heat produced in resistor R_1 after opening the switch k is

- a) $\frac{1}{2} \frac{LV^2}{R}$
b) $\frac{1}{2} \left(\frac{R_1}{R_1 + R_2} \right) \frac{LV^2}{R^2}$
c) $\frac{1}{2} \left(\frac{R_2}{R_1 + R_2} \right) \frac{LV^2}{R^2}$
d) zero

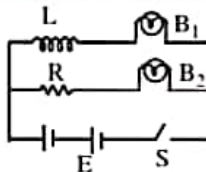


10. In the circuit shown, X is joined to Y for a long time, and then X is joined to Z. The total heat produced in R_2 is :

- a) $\frac{LE^2}{2R_1^2}$ b) $\frac{LE^2}{2R_2^2}$
 c) $\frac{LE^2}{2R_1R_2}$ d) $\frac{LE^2R_2}{2R_1^2}$



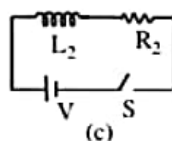
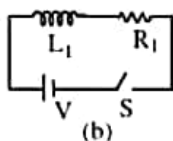
11. An inductor L , a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R . Which of the following statements gives the correct description of the happenings when the switch S is closed?



- a) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.
 b) B_1 light up earlier and finally both the bulbs acquire equal brightness.
 c) B_2 lights up earlier and finally B_1 shines brighter than B_2 .
 d) B_1 and B_2 light up together with equal brightness all the time.

More than One correct answer Type Questions

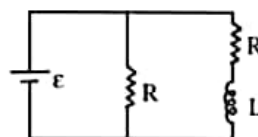
12. Current growth in two L-R circuits (b) and (c) are as shown in figure (a). Let L_1 , L_2 , R_1 and R_2 be the corresponding values in two circuits. Then



- a) $R_1 > R_2$ b) $R_1 = R_2$ c) $L_1 > L_2$ d) $L_1 < L_2$

13. In the circuit diagram shown

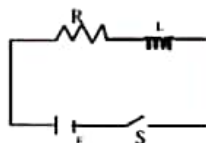
- a) time constant is L/R
 b) time constant is $2L/R$
 c) steady state current in inductor is $2\epsilon/R$
 d) steady state current in inductor is ϵ/R



Linked Comprehension Type Questions

Passage - I :

An LR circuit consists of an inductance L and a resistance R connected in series through a battery of emf E . At time $t = 0$, the switch S is closed.





14. Work done by the battery during the time $t = 0$ to $t = t$

- a) $\frac{E^2}{R} t e^{-\frac{Rt}{L}}$ b) $\frac{E^2}{R} t e^{-\frac{t}{RL}}$ c) $\frac{E^2}{R} \left\{ t - \frac{L}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \right\}$ d) $\frac{E^2}{R} \left\{ t + \frac{L}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \right\}$

15. The heat developed during this time is

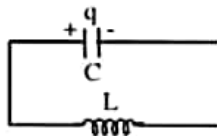
- a) $\frac{E^2}{R} \left\{ t - \frac{L}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \right\}$ b) $\frac{E^2}{R} \left\{ t - \frac{L}{2R} \left(3 - 4e^{-\frac{Rt}{L}} + e^{-\frac{2Rt}{L}} \right) \right\}$
c) $\frac{E^2}{R} \left\{ t + \frac{L}{R} \left(1 - e^{-\frac{Rt}{L}} \right) \right\}$ d) $\frac{E^2}{R} \left\{ t - \frac{L}{2R} \left(1 - 2e^{-\frac{Rt}{L}} + e^{-\frac{2Rt}{L}} \right) \right\}$

16. The magnetic field energy stored in the circuit at time t is

- a) $\frac{LE^2}{R^2} \left(1 - e^{-\frac{Rt}{L}} \right)^2$ b) $\frac{LE^2}{2R^2} \left(1 - e^{-\frac{Rt}{L}} \right)$ c) $\frac{LE^2}{2R^2} \left(1 + e^{-\frac{Rt}{L}} \right)^2$ d) $\frac{LE^2}{2R^2} \left(1 - e^{-\frac{Rt}{L}} \right)^2$

Passage - II :

In an L - C circuit shown in figure :



$C = 1\text{F}$, $L = 4\text{H}$. At time $t = 0$, charge in the capacitor is 4C and it is decreasing at a rate of $\sqrt{5}\text{C/s}$.

17. Maximum charge in the capacitor can be :

- a) 6C b) 8C c) 10C d) 12C

18. Charge in the capacitor will be maximum after time $t = \dots\dots\dots$ second

- a) $2\sin^{-1}\left(\frac{2}{3}\right)$ b) $\pi - 2\sin^{-1}\left(\frac{2}{3}\right)$ c) $2\tan^{-1}\left(\frac{2}{3}\right)$ d) none of these

19. Choose the correct option :

- a) Maximum current in the circuit is 4A
b) When current is half its maximum value, charge in capacitor is less than half its maximum value
c) both (a) and (b) are correct
d) both (a) and (b) are wrong



Matrix Matching Type Questions

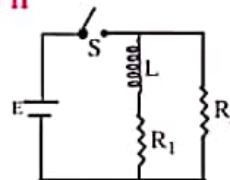
20. In the circuit shown in figure $E = 18\text{ V}$, $L = 2\text{ H}$, $R_1 = 3\Omega$, $R_2 = 6\Omega$, switch S is closed at $t = 0$. Match the following;

COLUMN - I

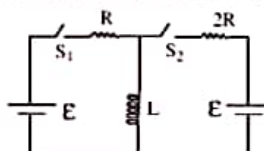
- A) Current through R_1 at $t = 0$
 B) Current through R_1 at $t = \infty$
 C) Current through R_2 at $t = 0$
 D) Current through R_2 at $t = \infty$

COLUMN - II

- p) 6 A
 q) 3 A
 r) Zero
 s) Infinite

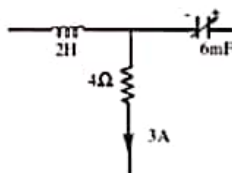
Integer Type Questions

21. In the circuit shown, switch S_2 remains closed for a long time, with S_1 open. It is given that $R = 10\Omega$, $L = 1\text{ mH}$ and $\mathcal{E} = 3\text{ V}$. Now switch S_1 is also closed. Immediately afterwards, if the magnitude of rate of change of current (in A s^{-1}) in the inductor is $1000x$, find x .



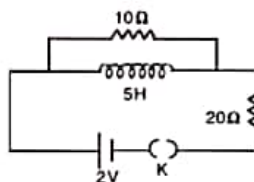
22. Figure shows a part of a bigger circuit. The capacity of the capacitor is 6 mF and is decreasing at the constant rate 0.5 mFs^{-1} . The potential difference across the capacitor at the shown moment is changing as follows :

$$\frac{dV}{dt} = 2\text{ Vs}^{-1}, \quad \frac{d^2V}{dt^2} = \frac{1}{2}\text{ Vs}^{-2}$$



The current in the 4Ω resistor is decreasing at the rate of 1 mA/s . What is the potential difference in (mV) across the inductor at this moment.

23. Two resistances of 10Ω and 20Ω and an ideal inductor of inductance 5 H are connected to a 2 V battery through a key K , as shown in figure. The key is inserted at $t = 0$. What is the final value of current in the 10Ω resistor ?



KEY SHEET (PRACTICE SHEET)

EXERCISE-I

LEVEL-I

1) 4 2) 4 3) 1 4) 4 5) 1.0 6) 6.0

LEVEL-II

1) a 2) d 3) a 4) a 5) c 6) b 7) c 8) d
 9) a 10) c 11) d 12) A-q,s; B-p,r; C-p,r; D-q,s
 13) A-r; B-p,q; C-s; D-r 14) A-p,s; B-p; C-p,s; D-q 15) 2 16) 1
 17) 2

EXERCISE-II

LEVEL-I

1) 3 2) 2 3) 2 4) 1 5) 2 6) 2.0 7) 7.0

LEVEL-II

1) b 2) c 3) a 4) b 5) a 6) a 7) a
 8) a,b,c,d 9) a,b 10) a,b,d 11) a 12) a,c,d 13) a,b,c
 14) b,c,d 15) b,d 16) c 17) d 18) b 19) c 20) d
 21) A-p,q; B-p,r; C-p,r; D-q 22) 8 23) 1 24) 5 25) 2
 26) 8

EXERCISE-III

LEVEL-II

1) a 2) c 3) a 4) b 5) d 6) a,b,c 7) b,d 8) d
 9) d 10) d 11) A-r; B-r; C-p,q 12) 3 13) 8 14) 2

EXERCISE-IV

LEVEL-I

1) 2 2) 2 3) 3 4) 0.16 5) 12.56

LEVEL-II

1) d 2) a 3) d 4) b 5) a,b,c,d 6) A-q; B-r; C-q
 7) 4

EXERCISE-V

LEVEL-I

1) 1 2) 2 3) 3 4) 1 5) 2 6) 4 7) 200.0 8) 888.8

LEVEL-II

1) c 2) d 3) d 4) b 5) c 6) a 7) b 8) c
 9) c 10) a 11) a 12) bd 13) ad 14) c 15) b 16) d
 17) a 18) b 19) d 20) A-p; B-r; C-q; D-q 21) 2 22) 4
 23) 0

ADDITIONAL PRACTICE EXERCISE

LEVEL-I (MAIN)

Straight Objective Type Questions

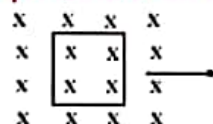
1. A square loop having n turns and an area A has a total resistance R . There is a uniform magnetic field ' B ' normal to plane of the loop as shown. If the loop is pulled out of the field slowly and uniformly in time t , work done in pulling it out completely.

1) $\frac{nB^2A^2}{Rt}$

2) $\frac{nB^2A}{Rt}$

3) $\frac{nBA^2}{Rt}$

4) $\frac{nB^2A^2}{R^2t}$



ELECTROMAGNETIC INDUCTION

LECTURE SHEET

EXERCISE-I

LEVEL-I (MAIN)

Straight Objective Type Questions

1. $\phi = BAN \cos \theta$
 $= 0.2 \times 5 \times 10^{-4} \times 100 \times \frac{1}{2} = 5 \times 10^{-3} \text{ wb}$
2. $\varepsilon = \frac{d\phi}{dt} \Rightarrow \varepsilon = \frac{BAN}{t}$
3. $\varepsilon = \frac{d\phi}{dt} = \frac{\phi_2 - (-\phi_1)}{t} = \frac{\phi_1 + \phi_2}{t}$
4. $\varepsilon = AN \left(\frac{dB}{dt} \right) \& i = \frac{\varepsilon}{R}$
5. $q = \frac{\Delta\phi}{R} = \frac{\phi_2 - \phi_1}{R} = \frac{B(A_2 - A_1)}{R}$

Numerical Type Questions

6. $\varepsilon_{\max} = BAN\omega$
7. Coil was rotated about vertical axis. Change in flux is zero.
8. $\varepsilon = AN \left(\frac{dB}{dt} \right)$
9. $q = \frac{\Delta\phi}{R} = \frac{BAN}{R}$
10. $E = \frac{d\phi}{dt}; \phi = BA = B_0 a^2 \rightarrow (1)$
 $E = \frac{d\phi}{dt} = B_0 a^2$
 Total resistance $R = \lambda \cdot 12a \rightarrow (2);$
 $i = \frac{E}{R} = \frac{B_0 a^2}{\lambda \cdot 12a}$

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. $R = R_0 + r^2; \varepsilon = \frac{d\phi}{dt} = \frac{d}{dt}(BA) = B \frac{d}{dt}(\pi R^2)$
 $= \pi B \frac{d}{dt}(R_0 + r^2)^2 = \pi B \cdot 2(R_0 + r^2) \cdot 2t$
 $= 4\pi Bt (R_0 + r^2)$

2. $O \rightarrow t_1 : B \text{ increases} \Rightarrow \text{current is ACW}$
 $t_1 \rightarrow t_2 : B \text{ is constant} \Rightarrow \text{current is zero}$
 $t_2 \rightarrow t_3 : B \text{ decreases} \Rightarrow \text{current is CW}$

$$3. E_{\text{avg}} = \frac{\phi_f - \phi_i}{\Delta t} = \frac{B \left(\frac{\pi r^2}{2} \right)}{\Delta t} = \frac{B \pi r^2}{2 \Delta t}$$

4. At any instant of time $\phi = \vec{B} \cdot \vec{A} = BA$

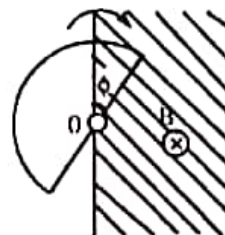
$$\left| \frac{d\phi}{dt} \right| = B \left(\frac{dA}{dt} \right)$$

$$\text{Area any instant : } l = [l_0 + l_0 \alpha (\theta_f - \theta_i)]^2$$

$$A = l^2; \theta_f - \theta_i = \omega t$$

$$\frac{dA}{dt} = 2l \left(\frac{dl}{dt} \right) \Rightarrow |E_{\text{ind}}| = 2B_0 l_0^2 \alpha b [1 + \alpha \omega t]$$

5. Area swept $\Lambda = \frac{1}{2} r^2 \phi \Rightarrow \Lambda = \frac{1}{2} a^2 \left[\frac{1}{2} \omega t^2 \right]$



$$\Rightarrow \frac{d\Lambda}{dt} = \frac{1}{4} a^2 \omega \cdot 2t = \frac{a^2 \omega t}{2}$$

$$\text{induced emf } E = B \cdot \frac{d\Lambda}{dt} = \frac{1}{2} \alpha B a^2 t$$

6. Magnetic field in the strip 'dx' due to the current $i = i_0 e^{-x/\lambda}$ is $B = \frac{\mu_0 i}{2\pi x}$

$$\text{flux } \phi = BA \Rightarrow \phi = \frac{\mu_0 i}{2\pi x} b \, dx$$

$$\Rightarrow \phi = \frac{\mu_0 i b}{2\pi} \int_d^{d+a} \frac{1}{x} dx = \frac{\mu_0}{2\pi} i b \ln \left[\frac{d+a}{d} \right]$$

$$\text{induced emf } \varepsilon = \frac{-d\phi}{dt} \Rightarrow \varepsilon = \frac{-\mu_0}{2\pi} i_0 b ;$$

$$\ln \left(\frac{d+a}{d} \right) \frac{d}{dt} (e^{-t/\tau}) \Rightarrow \varepsilon = \frac{\mu_0 i b}{2\pi \tau} \ln \left[\frac{d+a}{d} \right]$$

7. According to Lenz's law the magnetic field by current in AB is vertically upwards which is increasing. So the induced current in the loop will be clockwise direction so that its magnetic field is vertically downwards

$$8. dH = i^2 R dt; \rho = \frac{d\phi}{dt} = \frac{d}{dt}(6t - 3t^2)$$

$$\rho = 6 - 6t; i = \frac{\rho}{R}; dH = \frac{\rho^2}{R^2} R dt$$

$$dH = \frac{\rho^2}{R} dt; \int dH = \frac{1}{4} \int_0^2 (6 - 6t)^2 dt = 6J$$

More than One correct answer Type Questions

9. In Situation-I two magnets move towards each other with the same speeds. Hence the rate of flux is same and induced emfs are also same. Hence net current in the loop is zero.

In Situation-II with unequal speeds the magnets are approaching so the net emf in the loop is non-zero.

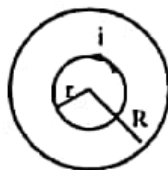
Linked Comprehension Type Questions

Passage - I :

$$10. \phi = BA \cos \theta$$

$$= \frac{\mu_0 n i}{2r} (\pi R^2 2n) \cos 60$$

$$= \frac{\mu_0 n^2 i \pi R^2}{2r}$$



$$11. q = \frac{d\phi}{K} = \frac{\mu_0 n^2 \pi R^2}{2rK} (i + i) = \frac{\mu_0 n^2 \pi R^2 i}{rK}$$

$$12. \text{Workdone}$$

$$W = MB (\cos \theta_1 - \cos \theta_2) = \frac{\mu_0 n^2 \pi i^2 r^2}{R}$$

Passage - II :

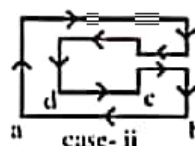
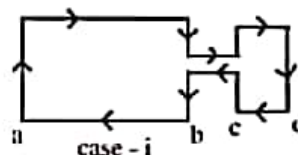
$$13. \phi = BA \text{ [Where 'A' is the effective area]}$$

$$\phi_1 = B(L^2 + l^2) \text{ and } \phi_2 = B(L^2 - l^2)$$

- (14-15) Case- i : As magnetic field constantly decreasing, induced current tries to increase the field. for this induced current produces an inward field. This is possible by flowing the current in bigger loop from b to a and in smaller loop d to c.

Case - ii : In second case induced currents are in reverse order for one to another loop. So for the same current one loop increases but another one decreases the induced field

To over come the decreasing rate we can select larger area loop as increasing field. To this current is from b to a



$$16. \epsilon = A \frac{dB}{dt} \quad E_2 < E_1 \Rightarrow i_2 < i_1$$

Matrix Matching Type Questions

17. If the magnetic flux linked with the rigid loop changes, by Lenz's law, to oppose the changing flux, an induced current and magnetic moment are developed in the loop. So the torque experienced by the loop is calculated by using, $\vec{\tau} = \vec{M}_{ind} \times \vec{B}_{app}$



A) If the current is increased, the flux through the loop increases along '-z' axis. By Lenz's law, to oppose the increasing flux, the induced current in the loop should be in ACW direction. Hence the induced magnetic moment is in the upward direction. $\vec{M}$ and $\vec{B}$ being anti parallel, there will be no torque on the loop. At the same time as the portion PQR is closer than PSR, magnetic repulsive force acts on the loop.

B) Similarly if current is decreased, the induced current is in cw direction and $\vec{M}$ and $\vec{B}$ are parallel. Hence the torque on the loop is zero. In this case there is an attractive force on the loop.

C) If the current carrying wire is moved away, the flux through the loop decreases. Hence the induced current in the loop is in cw direction. The torque is zero and an attractive force acts on the loop

D) Similar to 'C', if the wire is moved towards the loop, the flux through the loop increases. Hence the induced current in the loop is in ACW direction. In this case $\vec{M}$ and $\vec{B}$ being antiparallel the torque on the loop is zero. But a repulsive force acts on the loop.

18. A-R; when a metallic loop rotates about an axis parallel to the magnetic field.
 $BA \cos \theta = \text{constant} \Rightarrow \epsilon_{\text{ind}} = 0$
 B-P; Lenz's law explains the law of conservation of energy there by explained the cause for the opposition offered by a coil to the changing flux due to the work done by the external agent resulting in Joule's heat produced by the induced current in the coil.
 C-S; If the current in the loop decreases, the flux linked with the loop decreases resulting in induced emf opposing the change in flux linked with it
 D-Q; Flemming's right hand rule explains the direction of induced emf and hence induced magnetic field.

EXERCISE-II)

LEVEL-I (MAIN)

Straight Objective Type Questions

- $\epsilon = Bv\ell$
- $\epsilon = \frac{1}{2} R^2 \omega B$
- $\epsilon = \frac{1}{2} B r^2 \omega$ and it is independent of J number of spokes.
- $L_{\text{effective}} = 2r$, $\epsilon = B(2r)v$, $i = \frac{B(2r)v}{\left(\frac{R}{2}\right)}$

The total current in PQ is $i = i_1 + i_2 = \frac{8Brv}{R}$

- $V_1 = V \sin 30$, $V_A - V_B = B\ell(v \sin 30) = 4V$
- $mg \sin \theta = Bi\ell$;
 $mg \sin \theta = B \left(\frac{Bv}{R} \right) \ell \Rightarrow v = \frac{mgR \sin \theta}{B^2 \ell^2}$

Numerical Type Questions

- $\epsilon = B\ell v_0$;
 $i = \frac{\epsilon}{R} \Rightarrow 10^{-3} = \frac{0.2v}{4} \therefore v = 0.02 \text{ ms}^{-1}$
 (Balanced wheatstone bridge circuit)

$$8. \quad \epsilon = \vec{\ell} \cdot [\vec{v} \times \vec{B}] ;$$

$$\epsilon = 5\hat{j} \cdot [\hat{i} \times (3\hat{i} + 4\hat{j} + 5\hat{k})] = 25 \text{ volts}$$

9. **Method-I**

Length of the rod is $= 2x = 2\sqrt{\frac{y}{c}}$

Assume velocity of rod is 'v' and emf

$$= v(B)(2x) ; \epsilon = 2vB \sqrt{\frac{y}{c}}$$

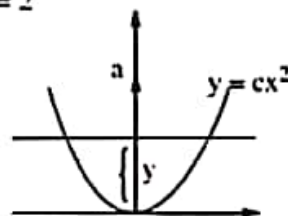
we know that $v^2 = u^2$

$$= 2as \Rightarrow v^2 = 2ay ; v = \sqrt{2ay}$$

$$\therefore \epsilon = 2\sqrt{2ay} B \sqrt{\frac{y}{c}} = By$$

$$\sqrt{\frac{8a}{c}} = 2By \sqrt{\frac{2a}{c}}$$

$$K = 2$$



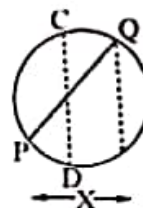
$$10. \quad P = vi = (Blv)i = 6.25 \text{ mw}$$

$$11. \quad -m \frac{dv}{dt} = \frac{B^2 \ell^2 v}{R} \Rightarrow -\int_{v_i}^v \frac{dv}{v} = \int_0^t \frac{B^2 \ell^2}{mR} dt$$

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- $BH = Bv \cot \theta = B \cot \theta$ $\epsilon = Blv \cot \theta$
- When the ring falls vertically there will be an induced emf across



A & B ($\epsilon = BV(2r)$). There will be a P.D across any two points on the ring of line joining then has projected length in the horizontal plane. Between points P and Q there is projected length 'X' in horizontal plane.

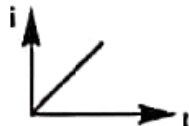
P.D across P and Q is $e = Bvx$

Therefore P.D = 0

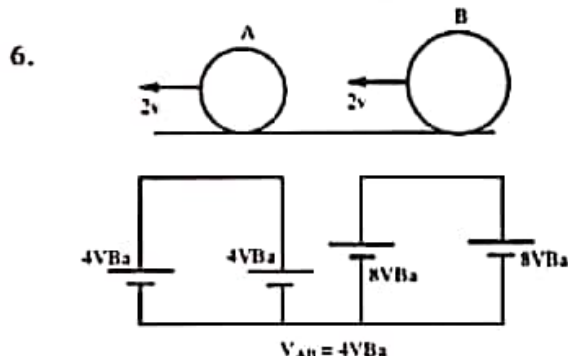
$$3. \quad e = \frac{d\phi}{dt} = \frac{BAN}{t}, \quad w = \frac{e^2}{R} \cdot t ;$$

$$w = \frac{B^2 A^2 N^2}{R t^2} \times t, \quad w = \frac{B^2 A^2 N^2}{R t}$$

4. Distance between the free ends increases as $y = mx$



$$5. \quad \text{induced emf } E = B/\partial = \frac{\mu_0 i}{2\pi r}$$



7. As long as the square loop is inside the magnetic field, $\text{emf} = 0$ and hence $i = 0$. Therefore the force and hence work done is zero.

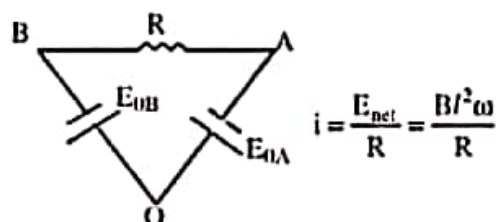
$$8. \quad dE = B \partial dx \Rightarrow \int dE = \int_a^b \frac{\mu_0 i_0}{2\pi x} \partial dx$$

$$\Rightarrow E = \frac{\mu_0 i_0}{2\pi} \partial \ln x \Big|_a^b$$

$$\Rightarrow E = \frac{\mu_0 i_0}{2\pi} \partial \ln(b/a) \quad \text{Power } P = F \partial$$

$$F = \frac{P}{\partial} = \frac{E^2}{R \partial} \Rightarrow F = \left[\frac{\mu_0 i_0^2}{2\pi} \ln(b/a) \right] \frac{\partial}{R}$$

$$9. \quad E_{0A} = \frac{1}{2} B l^2 \omega \quad \text{and} \quad E_{0B} = \frac{1}{2} B l^2 \omega$$



$$10. \quad dE = B \partial dx$$

$$\Rightarrow dE = B \omega x dx$$

$$\int dE = B \omega \int_{2l}^{3l} x dx$$

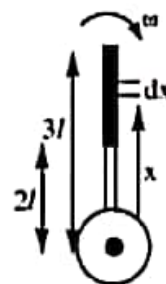
$$\Rightarrow E_{in} = \frac{B \omega}{2} [9l^2 - 4l^2]$$

$$= \frac{5}{2} B l^2 \omega$$

E_{in} for non coating material = zero

$\therefore E_{in}$ for the total length of the rod is only

$$\text{due to copper -coating length} = \frac{5}{2} B l^2 \omega$$



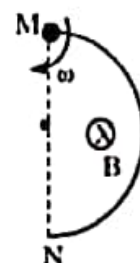
11. effective length between M and N is diameter of semicircle

$$x = 2(L/\pi)$$

$$E_{MN} = \frac{1}{2} B x^2 \omega$$

$$L = \pi r \Rightarrow r = L/\pi$$

$$= \frac{1}{2} B \omega \frac{4L^2}{\pi^2} = \frac{2B\omega L^2}{\pi^2}$$



12. The induced emf across the centre and any point on the circumference is

$$|e| = \frac{B \omega r^2}{2}; \quad i = \frac{e}{R} = \frac{B \omega r^2}{2R}$$

More than One correct answer Type Questions

13. Rate of workdone by external agent is

$$\frac{dW}{dt} = \frac{B i L dx}{dt} = B i v \quad \& \quad \text{thermal power}$$

dissipated in the resistor $= (Bv/i) i$ clearly both are equal.

If applied external force is doubled, the rod will experience a net force and hence acceleration as a result velocity increases

$I = \frac{e}{R}$ on doubling 'R' current and hence required power becomes half. $P = B i v$

14. The initial kinetic energy $= \frac{1}{2} m v_0^2$: Final K.E = 0

Change in K.E. = $\frac{1}{2} m v_0^2 = \text{work done}$

Force on the conductor $dF = iBdl$

where $i = \frac{Bv_0 l}{R} \therefore dF = \frac{B^2 v_0 l}{R} dl$

The average force is given by

$$F = \int_0^l \frac{B^2 v_0 l}{R} dl = \frac{B^2 v_0 l^2}{2R}$$

Now work done = Average force $\times$ distance

$$\frac{1}{2} m v_0^2 = \frac{B^2 v_0 l^2}{2R} \times S \Rightarrow S = \frac{m v_0 R}{B^2 l^2}$$

Also, Heat generated = loss of K.E. = $\frac{1}{2} m v_0^2$

15. a) In the diagram, PQ rod doesn't cut the magnetic field. So there is no current passing through PQ. So potential difference across PQ is zero

d) The potential difference across QR is emf = $B_0(a\omega)(a) = B_0 \omega a^2$.

16. $\phi = BA = B_0 \pi r_0^2 t^3$; $E = \frac{d\phi}{dt} = B_0 \pi r_0^2 3t^2$

$E \propto t^2$. (parabola) $\rightarrow$ 'B' option is true

$\frac{B_0}{r_0} = ML^{-1}T^{-2}A^{-1} \rightarrow$ 'C' option is true

$$\frac{dE}{dt} = \frac{d(t^2)}{dt} = 2t \therefore \frac{dE}{dt} \propto t$$

$\rightarrow$ 'A' option is true

Linked Comprehension Type Questions

Passage :

(17 to 18) : The charge on the capacitor decreases and the voltage across it also decrease. The voltage induced in the rod increases until the two voltages cancel each other.

$$B/V_{\max} = \frac{Q_{\min}}{C} ; m \cdot \frac{dV}{dt} = -B \frac{dQ}{dt}$$

$$mV_{\max} = B(Q_0 - Q_{\min})$$

$$V_{\max} = \frac{B/CV_0}{m+B^2l^2C} \text{ and } Q_{\min} = \frac{B^2l^2C^2V_0}{m+B^2l^2C}$$

19. The heat generated in the system is the difference between the initial and final energies

Matrix Matching Type Questions

20. For Case -I, $V_B - V_A = B_0 \frac{\omega l^2}{2} = B_0 v l$

$$\left(\because w = \frac{2v}{l} \right)$$

Integer Type Questions

21. Potential difference between O and A is Blv
Where v is the velocity of the mid-point of O and A

$$\text{Thus, } e = B_0 \left(\frac{L}{2} \right) \left[v_0 + \frac{L}{4} \left(\frac{v_0}{L} \right) \right] = \frac{5B_0 L v_0}{8}$$

22. conducting surface $V_A = V_B = 0$

$$E_{AC} = \frac{1}{2} B \omega l^2$$

$$\Rightarrow E_A - E_C = -\frac{1}{2} B \omega 4r^2$$

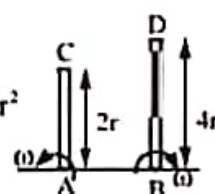
$$E_C = -2B\omega r^2$$

$$\Rightarrow E_C = 4Bv r \quad [2V = r\omega]$$

$$E_{BD} = -\frac{1}{2} B \omega (16r^2)$$

$$\Rightarrow E_B - E_D = -8B\omega r r \Rightarrow E_D = 8Bv r$$

$$\therefore E_{DC} = E_D - E_C = 4Bv r$$



EXERCISE-III

LEVEL-II (ADVANCED)

Straight Objective Type Questions

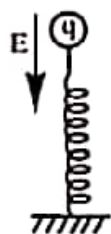
- $E \propto \frac{1}{R}$
- $\oint \vec{E} \cdot d\vec{r} = -\frac{d\phi}{dt}$ and take the sign of flux according to right hand curl rule
 $\oint \vec{E} \cdot d\vec{r} = -(-\alpha A) - (-\alpha A) + (-\alpha A) = -\alpha A$
- Torque $\tau = I\alpha \Rightarrow E q R = mR^2 \alpha$
 $\left(\frac{1}{2} R \frac{dB}{dt} \right) q R = mR^2 \alpha \Rightarrow \alpha = \frac{q}{2m} \frac{dB}{dt}$
- zero, since if any induced current passing through AB then, in half of the circle magnetic field due to induced current increases and at the same rate magnetic field decreases in the remaining circle.

∴ net field by this is zero, so it does not oppose the change of flux.

5. Flux in the frame $\phi = BA = B \frac{\pi r^2}{2}$
induced current $i = \frac{1}{R} \frac{d\phi}{dt} = \frac{1}{R} \frac{\pi r^2}{2} \frac{dB}{dt}$

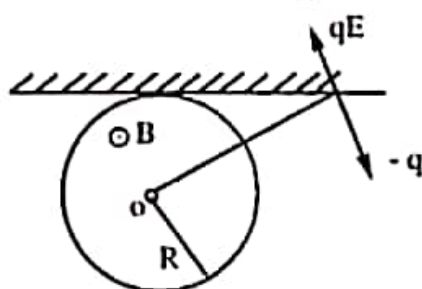
6. $E = \frac{1}{2} r^2 \left(\frac{dB}{dt} \right)$
 $Eq + mg = kx$

$x = \frac{1}{k} \left[mg + \frac{qr^2}{2l} \frac{dB}{dt} \right]$



7. Let $\mathcal{E}$ to be induced emf in loop = $\frac{C_1 \pi r^2}{6}$
 V is the potential drop across wire AB
 $\mathcal{E} - V = iR \rightarrow (1) ; V = 2iR ; V = \frac{2}{3} \mathcal{E} ;$
 $V = \frac{C_1 \pi r^2}{9}$. The direction is reversed.

8. According to $\int \vec{E} \cdot d\vec{l} = \frac{d\phi}{dt}$



$E = \frac{R^2 K}{2l}$

$Eq \cos \theta$ is balanced by normal force.

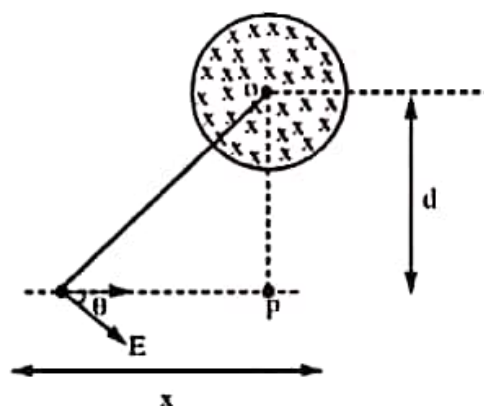
Net force is $Eq \sin \theta = \frac{qkR^3}{2l^2}$

More than One correct answer Type Questions

9. For $r \leq R$, $E(2\pi r) = \frac{dB}{dt} \pi r^2$, $E = \frac{\alpha r}{2}$
For $r > R$, $E(2\pi r) = \frac{dB}{dt} (\pi R^2)$, $E = \frac{\alpha R^2}{2r}$
For $r = \frac{R}{2}$, the emf induced,
 $\mathcal{E} = \frac{dB}{dt} \left(\pi \left(\frac{R}{2} \right)^2 \right) = \frac{\pi \alpha R^2}{4}$

10. At a distance 'r' from 'P' on a line perpendicular to QP, the induced electric field would be

$E 2\pi \sqrt{x^2 + d^2} = \pi R^2 \frac{dB}{dt}$



$\Rightarrow E = \frac{KR^2}{2} \times \frac{1}{\sqrt{x^2 + d^2}} \left[\because \frac{dB}{dt} = K \right]$

The force experienced by charged particle due to induced electric field in $F = -qE$. The component of this electric force along the line perpendicular to OP is

$F_x = qE \cos \theta = Q \frac{kR^2}{2} \frac{d}{x^2 + d^2}$

Using work energy theorem,

$\Delta K = W_{\text{external agent}} + W_f ;$

$0 = W_{\text{external agent}} + \int -f_x \cdot dx$

$W_{\text{external agent}} = \int_0^{\infty} f_x \cdot dx = \frac{qkR^2 d}{2} \int_0^{\infty} \frac{dx}{x^2 + d^2}$

$= \frac{q\pi R^2}{4} K$. This value doesn't change with 'd'.

Linked Comprehension Type Questions

Passage - I :

11. $E = \frac{r}{2} \frac{dB}{dt}$

$\frac{dB}{dt} = B_0 \Rightarrow E = \frac{n}{2} B_0$

$d\tau = d\vec{l} \cdot \vec{a} = (dqE) \cdot \vec{a}$



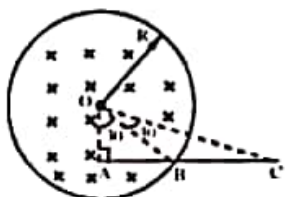
$$\int d\tau = E \int dq = Eaq ; \tau = I\alpha$$

$$\alpha = \frac{\tau}{I} = \frac{Eaq}{(ma^2)} = \frac{Eq}{ma} \text{ anticlockwise}$$

$$12. P = \tau \cdot w = (Eaq)(\alpha) = Eaq \frac{Eq}{ma} \quad P = \frac{E^2 q^2}{m}$$

Passage - II :

$$13. e = \frac{d\phi}{dt} = A \frac{dB}{dt} = (\text{Area of } OAB \Delta^{ic}) \alpha$$



$$= \left[\frac{1}{2} L (AB) \right] \alpha = \left[\frac{1}{2} L (L \tan 30) \right] \alpha = \frac{L^2 \alpha}{2\sqrt{3}}$$

$$14. e_{BC} = \left[\text{Area of } OBC \Delta^{ic} \right] \frac{dB}{dt} = \left[\frac{\pi R^2}{360} \times 30 \right] \alpha$$

$$= \frac{\pi R^2}{12} \alpha$$



$$15. \text{ from } dv = -E dr \quad -\frac{dB}{dt} A = -E 2\pi (2L)$$

$$\Rightarrow \pi R^2 \alpha = E 4\pi L \Rightarrow E = \frac{R^2 \alpha}{4L}$$

EXERCISE-IV

LEVEL-I (MAIN)

Straight Objective Type Questions

$$1. e = L \frac{di}{dt}$$

$$2. \phi_s = Mi_p$$

$$3. U = \frac{1}{2} Li^2$$

Numerical Type Questions

4. Mutual inductance is for pair of coils in both cases same because total number of turns are same.

$$5. \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$$

$$6. B = \mu_0 ni = \frac{\mu_0 \theta_1 i}{L} ; \phi = BA \theta_2 = \frac{\mu_0 \theta_1 i}{L} \cdot A \cdot N_2$$

$$\therefore M = \frac{\mu_0 \theta_1 N_2 A}{L}$$

$$M = \frac{4\pi \times 10^{-7} \times 300 \times 400 \times 10 \times 10^{-4}}{20 \times 10^{-2}}$$

$$M = 7.54 \times 10^{-4} \text{ H}$$

$\therefore$ value of 'X' is 7.54

$$7. U = \frac{1}{2} Li^2 \Rightarrow \frac{du}{dt} = Li \left(\frac{di}{dt} \right)$$

LEVEL-II (ADVANCED)

Straight Objective Type Questions

$$1. \phi_{\text{secondary}} = M i_{\text{primary}}$$

$$BA \cos 30 = M i_p \quad \frac{\mu_0 i_p \pi b^2}{2a} \times \frac{\sqrt{3}}{2} = M i_p$$

$$M = \frac{\sqrt{3} \mu_0 \pi b^2}{4a}$$

$$2. \phi = \int B \cdot dA = \int \frac{\mu_0 i}{2\pi r} \cdot b \cdot dr = \frac{\mu_0 i b}{2\pi} \int \frac{1}{r} \cdot dr$$

$$\phi = \frac{\mu_0 i b}{2\pi} \ln \left[\frac{x+a}{x} \right]$$

so, mutual inductance between rectangular loop and the long straight wire is $\phi = Mi$

$$\frac{\mu_0 i b}{2\pi} \ln \left(\frac{x+a}{x} \right) = Mi \Rightarrow M = \frac{\mu_0 b}{2\pi} \ln \left(\frac{x+a}{x} \right)$$

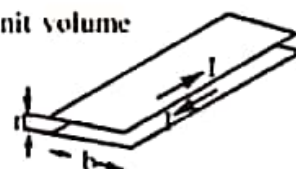
$$*3. \text{ field between plates } = \frac{\mu_0 i}{2} + \frac{\mu_0 i}{2} = \mu_0 i$$

$$B = \mu_0 i = \frac{\mu_0 I}{b} \quad (\because i = I/b)$$

Energy stored perunit volume

$$= \frac{B^2}{2\mu_0} = \frac{\mu_0^2 I^2}{2\mu_0 b^2}$$

$$\text{Energy} = \frac{\mu_0 I^2}{2b^2} (\text{volume})$$



$$= \frac{\mu_0 I^2}{2b^2} (b h) \Rightarrow \frac{1}{2} L i^2 = \frac{\mu_0 I^2}{2b} h$$

$$\Rightarrow L = \frac{\mu_0 h I}{b} \Rightarrow \frac{L}{I} = \frac{\mu_0 h}{b}$$

$$4. \phi = Li \Rightarrow BA = Li \Rightarrow i = \frac{B \pi a^2}{L}$$

$$5. W = U_f - U_i = \frac{1}{2} L (i_f^2 - i_i^2) = \frac{1}{2} L \left(\frac{B \pi a^2}{L} \right)^2$$

$$= \frac{B^2 \pi^2 a^4}{2L}$$

- *6. Within a cylindrical shell of length l , radius r and thickness dr , the magnetic field

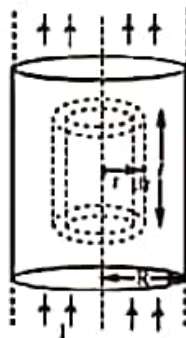
$B = \frac{\mu_0 I}{2\pi R^2} r$. This much uniform magnitude field exists within this shell. Energy stored in this shell is

$dU = (\text{energy density}) (\text{volume})$

$$= \left(\frac{B^2}{2\mu_0} \right) (dv)$$

$$= \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi R^2} r \right)^2 2\pi r dr l$$

$$= \frac{\mu_0 I^2 l}{2\pi R^4} r^3 dr$$



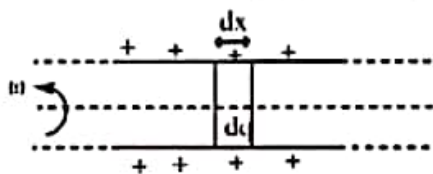
Total magnetic energy in a given wire of length ' l ' is

$$U = \int du = \frac{\mu_0 I^2 l}{4\pi R^4} \int_0^R r^3 dr = \frac{\mu_0 I^2 l}{4\pi R^4} \frac{R^4}{4} = \frac{\mu_0 I^2 l}{16\pi}$$

$$\text{Energy stored per unit length} = \frac{U}{l} = \frac{\mu_0 I^2}{16\pi}$$

- *7. The given charged cylinder is equivalent to a current carrying long solenoid.

$$dq = \lambda dx ; dI = (dq) \frac{\omega}{2\pi} = \lambda (dx) \frac{\omega}{2\pi}$$



$$\text{Current per unit length} \frac{dI}{dx} = \frac{\lambda \omega}{2\pi}$$

From the concepts of solenoid, at any inside point of solenoid, $B = \mu_0 nI$ and it is uniform. Here nI is equal to current per unit length along

$$\text{the length} \therefore B = \mu_0 \frac{dI}{dx} = \frac{\mu_0 \lambda \omega}{2\pi}$$

Energy stored per unit length of given cylinder

$$U = \left(\frac{B^2}{2\mu_0} \right) \frac{(\text{volume})}{\text{length}} = \frac{B^2}{2\mu_0} \times \pi a^2$$

$$= \frac{\pi a^2}{2\mu_0} \left[\frac{\mu_0 \lambda \omega}{2\pi} \right]^2 = \frac{\mu_0 \lambda^2 \omega^2 a^2}{8\pi}$$

$$8. i = \frac{q}{t} = \frac{q\omega}{2\pi}$$

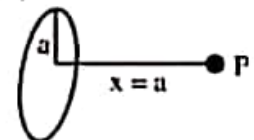
Ratio of volume energy densities at P

$$\frac{X_B}{X_E} = \frac{\frac{B^2}{2\mu_0}}{\frac{1}{2} \epsilon_0 E^2} = \frac{B^2}{\mu_0 \epsilon_0 E^2} \dots \dots \dots (1)$$

$$\text{at 'P', } B = \frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}} \text{ and}$$

$$E = \frac{1}{4\pi \epsilon_0} \frac{qx}{(r^2 + x^2)^{3/2}}$$

$$\frac{B}{E} = \frac{2\pi \epsilon_0 \mu_0 i r^2}{qx}$$



$$\Rightarrow \frac{B}{E} = \frac{2\pi \epsilon_0 \mu_0 q \omega}{2\pi} \frac{r^2}{qx} = \frac{r^2}{C^2 x}$$

$$\left[\because C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \right]$$

$$\therefore \text{from (1)} \frac{X_B}{X_E} = \left(\frac{B}{E} \right)^2 \frac{1}{\mu_0 \epsilon_0} = \frac{r^4}{C^4 x^2} C^2$$

$$= \frac{r^4}{C^2 x^2} = 1.1 \times 10^{-15}$$

Linked Comprehension Type Questions

Passage :

In stantainuous charge on capacitor is

$$q = CV_C = 6e^{-2t} ; I_C = \frac{dq}{dt} = -12e^{-2t}$$

'q' is decreasing with time.

$\therefore$ charge on the plates (+q and -q) is decreas-

ing. +ve charge is going from junction (from 'O') to the lower plate of the capacitor.
 $\therefore I_C$ is always away from 'O'

9. At junction 'O' : $I_1 + I_2 + I_L - I_C = 0 \Rightarrow I_L$
 $= I_C - I_1 - I_2 = 12e^{-2t} - e^{-2t} - 4 = 11e^{-2t} - 4$

10. $V_L = -L \frac{di}{dt} = -4[11e^{-2t}(-2) - 0] = 88e^{-2t}$

11. From loop theorem between P and Q
 $V_P - i_1 R_1 + i_2 R_2 - V_Q = 0$
 $\Rightarrow V_P - V_Q = i_1 R_1 - i_2 R_2 = 2e^{-2t} - 12$

12. $V_P - i_1 R_1 + \frac{q}{C} - V_R = 0$
 $\Rightarrow V_P - V_R = i_1 R_1 - \frac{q}{C} = 2e^{-2t} - \frac{6e^{-2t}}{2} = -e^{-2t}$

EXERCISE-V

LEVEL-I (MAIN)

Straight Objective Type Questions

1. $i = i_0 \left[1 - e^{-\frac{Rt}{L}} \right] \Rightarrow i = [1 - e^{-1}]$

2. $i = i_0 \left[1 - e^{-\frac{Rt}{L}} \right]$ & $V_L = -L \frac{di}{dt}$

Numerical Type Questions

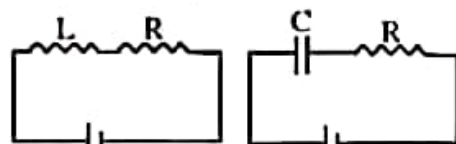
3. $U = \frac{1}{2} Li^2 \Rightarrow U \propto i^2$ $U = \frac{U_0}{4} \Rightarrow i = \frac{i_0}{2}$

from $i = i_0 e^{-t/\lambda} \Rightarrow \frac{i_0}{2} = i_0 e^{-t/\lambda} \Rightarrow t = \lambda \ln 2$

Charge flowing through the circuit is $q = \int idt$

$\Rightarrow q = \int_0^t i_0 e^{-\frac{Rt}{L}} dt = i_0 \left[\frac{e^{-t/\lambda}}{(-1/\lambda)} \right]_0^t = \frac{Li_0}{2R}$

4. After switch is closed,
 AB and CD are parallel to the battery



after a time 't'

after a time 't'

$i_2 = i_0 \left[1 - e^{-\frac{Rt}{L}} \right] = \frac{V}{R} \left[1 - e^{-\frac{Rt}{L}} \right]$

$q_c = q_0 \left[1 - e^{-\frac{t}{RC}} \right]$

$i_c = \frac{dq_c}{dt} = \frac{q_0}{RC} e^{-\frac{t}{RC}} = \frac{V}{R} e^{-\frac{t}{RC}}$

$\Rightarrow i_L = i_c \Rightarrow t = RC \ln 2$

5. $V_A - iR - \frac{Ldi}{dt} + E = V_B$

6. $U = \frac{1}{2} Li^2 = \frac{1}{2} Li_0^2 \left[1 - e^{-\frac{Rt}{L}} \right]^2$

7. $\frac{Q^2}{2C} = 2 \left[\frac{q^2}{2C} \right]$

8. $i_1 = i_0 \left[1 - e^{-\frac{Rt}{L}} \right]$; $i_2 = i_0 \left[1 - e^{-\frac{t}{CR}} \right]$

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. $i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right)$

$e = L \frac{di}{dt} = L i_0 \left[0 - e^{-\frac{Rt}{L}} \cdot \left(\frac{-R}{L} \right) \right] = E e^{-Rt/L}$

$\Rightarrow e^{-Rt/L} = e/E$

$\therefore i = i_0 \left[1 - \frac{e}{E} \right]$

$i = -\frac{i_0}{E} e + i_0$

$\Rightarrow i = -\frac{1}{R} e + i_0 \Rightarrow e = -Ri + Ri_0$

$y = -mx + c$

2. $i = 10e^{-4t}$ $e = -L \frac{di}{dt}$

